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Paper 39
Date: July 9, 2025

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

GENEOSCOPY, INC.,
Petitioner,

v.

EXACT SCIENCES CORPORATION,
Patent Owner.

IPR2024-00459
Patent 11,634,781 B2

Before RYAN H. FLAX, DAVID COTTA, and JAMIE T. WISZ, *Administrative Patent Judges.*

FLAX, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Exact Sciences Corporation ("Patent Owner") is the owner of U.S. Patent 11,634,781 B2 (Ex. 1001, "the '781 Patent"). Paper 4, 1 (Patent Owner's Mandatory Notices). On January 11, 2024, Geneoscopy, Inc. ("Petitioner") filed a Petition for *inter partes* review challenging the patentability of claims 1–20 of the '781 patent (all issued claims). Paper 1, 12 ("Pet."). We instituted trial in this proceeding on July 26, 2024. Paper 9 ("Institution Decision" or "DI").

Patent Owner filed a Response to the Petition (and the Institution Decision) on October 25, 2024. Paper 20 ("Response" or "Resp."). On January 27, 2025, Petitioner filed a Reply (Paper 27, "Reply") to Patent Owner's Response, which was followed by Patent Owner filing a Sur-reply (Paper 30, "Sur-reply") on March 10, 2025.

Oral argument was heard at a hearing held on May 1, 2025, and a transcript of the hearing is of record. *See* Paper 38 ("Hr'g Tr.").

Petitioner bears the burden of proving unpatentability of the challenged claims, and this ultimate burden of persuasion never shifts to Patent Owner. *Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). To prevail, Petitioner must prove unpatentability by a preponderance of the evidence. *See* 35 U.S.C. § 316(e) (2018); 37 C.F.R. § 42.1(d) (2023). After considering the parties' arguments and supporting evidence, as discussed in detail below, we conclude that Petitioner proves by a preponderance of the evidence that the challenged claims of the '781 patent are unpatentable.

A. Real Parties-in-Interest

Petitioner identifies itself, "Geneoscopy, Inc.," as the real party-in-interest. Pet. 2. Patent Owner identifies itself, "Exact Sciences Corporation," as the real party-in-interest. Paper 4, 2.

B. Related Matters

The parties identify *Exact Sciences Corporation v. Geneoscopy, Inc.*, No. 23-cv-1319-MN (D. Del.) as involving the '781 Patent and to be a related matter. Pet. 2–3; Paper 4, 2.

C. The '781 Patent and Relevant Background

The '781 Patent, titled "Fecal Sample Processing and Analysis Comprising Detection of Blood,"[1] was filed as U.S. Application 17/936,335 on September 28, 2022, and indicates priority to a series of continuation applications, including U.S. Application 16/634,607 ("the '607 Application"), and Provisional Application 61/149,581 ("the '581 Provisional"), which was filed on February 3, 2009. Ex. 1001, codes (54), (21), (22), (60), (63), 1:8–19. Thus, the earliest possible effective filing date of the '781 Patent is February 3, 2009. There is no dispute in this proceeding that the '781 patent is entitled to this priority date and we apply this priority date in our analysis here. *See* Pet. 4; *see generally* Resp.

The '781 Patent states that it relates to:

> A method of processing a fecal sample from a human subject comprising removing a portion of a collected fecal sample and adding the removed portion of the sample to a buffer that prevents denaturation or degradation of blood proteins found in

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[1] The '781 patent uses both "fecal" and "faecal," which refer to the same thing (solid body waste), but the former is the American spelling of the term and the latter is the British spelling. We use the American spelling herein.

> the sample, and detecting the presence of human blood in the removed portion of the fecal sample. The method further comprises stabilizing the remaining portion of the fecal sample.

Ex. 1001, Abstract, 1:30–31. According to the Specification, colorectal cancer ("CRC") is a leading cause of cancer-related deaths. *Id.* at 1:41–42. Most colon cancers arise from adenomatous polyps, which are usually asymptomatic. *Id.* at 1:46–52. Mass screening of asymptomatic patients is described as the cornerstone for detecting and eliminating these precursor lesions to reduce the risk of CRC. *Id.* at 1:52–55.

Colonoscopy is described as the primary screening test for CRC because of its high sensitivity and specificity and the ability to remove polyps if found. *Id.* at 1:65–2:1. The procedure, however, is invasive, costly, and has certain risks, such as infection and bowel perforation. *Id.* at 2:1–3. Fecal occult blood testing ("FOBT"), which tests for blood in the stool, is described as commonly used, and less invasive and less expensive than colonoscopy. *Id.* at 2:4–12. But, because occult blood in stool can be indicative of different gastrointestinal disorders, further testing is necessary to detect CRC. *Id.* at 2:9–12. The Specification describes two types of FOBT: guaiac FOBT ("gFOBT"), which detects peroxidase activity of hemoglobin in fecal blood, and immunochemical FOBT ("iFOBT" or "FIT"), which uses anti-human hemoglobin (Hb) antibodies to detect fecal blood. *Id.* at 2:13–34. Although iFOBT is more complicated and more expensive, it is more sensitive than gFOBT. *Id.* at 2:25–40.

The Specification also describes that recent developments in testing look specifically for mutations in DNA that are characteristic of colorectal neoplasia and are detectable in exfoliated epithelial cells in the stool. *Id.* at 2:44–47. The Specification describes that increased DNA methylation is an epigenetic alteration that is common in human cancers. *Id.* at 3:5–7. Aberrantly methylated DNA has also been proposed as a tumor marker for CRC detection. *Id.* at 3:7–9.

The ’781 patent further describes that, although combined assays for detecting CRC have been described, their approach targets either multiple protein markers or multiple DNA alterations. *Id.* at 3:41–43. According to the Specification, “[t]o date, immunochemical tests and DNA tests for CRC detection have been evaluated and compared on a separate basis only.” *Id.* at 3:43–45. The ’781 patent states that the invention “aims to improve the positive and negative predictive value and also the sensitivity and specificity of detection of colorectal cancer through non-invasive means.” *Id.* at 6:42–45. Accordingly, the invention is based upon a combination of tests for detecting proteins and epigenetic modification markers in the same fecal sample. *Id.* at 6:49–53.

The Specification describes that protein detection occurs in a step of “detecting the presence of blood in the f[]ecal sample,” “through any suitable means,” including “chromogenic or immunological” techniques, and the detection of an epigenetic modification is “detected in at least one gene selected from PHACTR3, NDRG4, FOXE1, GATA4, GPNMB, TFPI2, SOX17, SYNE1, LAMA1, MMP2, OSMR, SFRP2 and CDO1.” *Id.* at 7:8–24, 8:42–54. The Specification states that “‘[e]pigenetic modification’ is defined herein to include any alteration in the DNA, generally resulting in diminished gene expression, which is mediated by mechanisms other than alterations in the primary nucleotide sequence of a gene.” *Id.* at 12:35–39.

The Specification states that “[t]he inventor has determined that f[]ecal samples for use in the invention do not necessarily have to be freshly collected" and only "a minimal stool sample . . . may successfully be employed in the methods of the invention." *Id.* at 11:38–39, 11:55–58. The Specification describes that "a single f[]ecal sample is utili[z]ed as the source of the sample for" the two detection tests noted above, "[h]owever, the sample may need to be split to permit [this]." *Id.* 17:57–60. "Thus the sample may need to be treated to differing degrees in order to permit the separate steps to be carried out" and "[a]s a result, after obtaining the f[]ecal sample, the sample may be apportioned appropriately to allow the DNA and protein testing to be carried out separately." *Id.* at 17:60–64.

Once this portioning of the sample occurs, the Specification describes, one portion is combined with "a buffer which prevents denaturation or degradation of blood proteins," such buffers being known and commercially available, and another portion of the sample may be combined with "[s]uitable buffers for maintaining DNA integrity." *Id.* at 18:1–28. The Specification states that a "homogenization buffer" may be included and that such "are known in the art and commercially available," but no specific DNA stabilizing buffer is named. *Id.* at 27:44–55; *see generally id.*

The '781 patent concludes with twenty claims, of which claim 1 is the sole independent claim; it is illustrative of the challenged claims and we reproduce it below:

> 1. A method of processing a freshly-collected fecal sample without freezing, the method comprising:
>
> a) collecting a fecal sample from a human subject, wherein the fecal sample is collected at home by the human subject by defecation directly into a sealable collection vessel;
>
> b) removing a portion of the fecal sample to a separate sealable container to produce a removed portion and a remaining portion of the fecal sample;

> c) combining the removed portion of the fecal sample in the separate sealable container with a buffer that prevents denaturation or degradation of blood proteins found in a fecal sample, and sealing the sealable container; and
>
> d) combining the remaining portion of the fecal sample in the sealable collection vessel with a stabilizing buffer, and sealing the sealable collection vessel.

Ex. 1001, 45:21–47:4 (claims).

For context, according to the evidence of record, the ’781 patent is, as of 2021, one of 59 U.S. patents Patent Owner indicates cover its Cologuard product, approved for use by the U.S. FDA on August 11, 2014. Ex. 1084, 1; Ex. 2036, 5. As of its annual 10-K report in 2014 introducing this Cologuard product, Patent Owner identified that, at that point, it had "intellectual property rights pertaining to sample type, sample preparation, sample preservation, biomarkers, and related methods and formulations," protected then by 12 issued U.S. patents, as well as (potentially) 30 pending U.S. patent applications. Ex. 2036, 11.

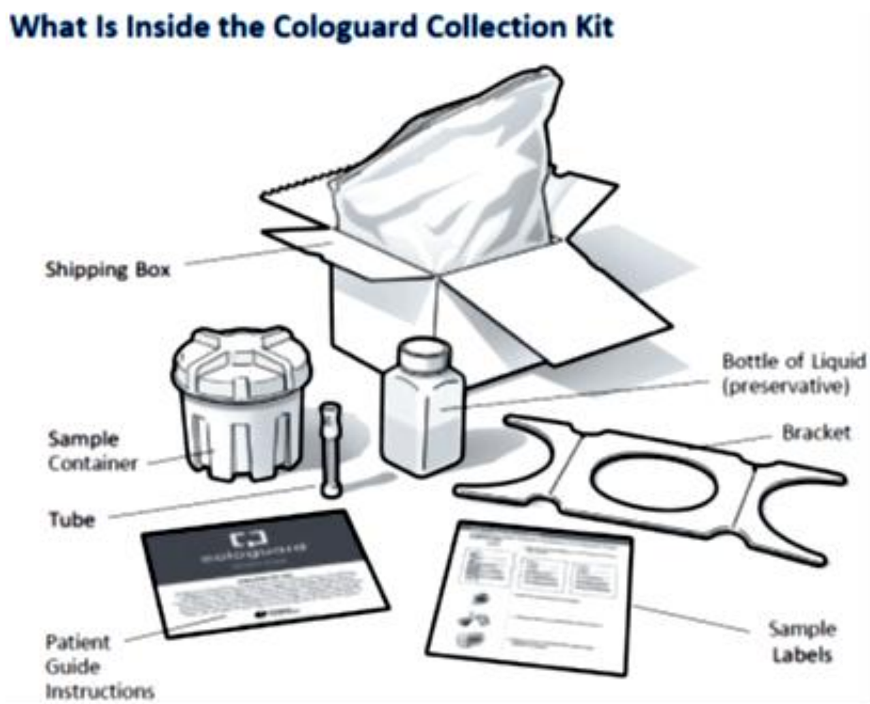
According to Patent Owner's published "Instruction for Use" for the Cologuard product,

> Cologuard is intended for the qualitative detection of colorectal neoplasia associated DNA markers and for the presence of occult hemoglobin in human stool. Cologuard is for use with the Cologuard collection kit and the following instruments: BioTek ELx808 Absorbance Microplate Reader; Applied Biosystems® 7500 Fast Dx Real-Time PCR; Hamilton Microlab® STARlet; and the Exact Sciences System Software with Cologuard Test Definition.

Ex. 1082, 6. "Cologuard is designed to analyze patients' stool for the presence of 11 molecular markers, including hemoglobin and DNA markers, which may indicate the presence of colorectal cancer or advanced adenomas.

Because cellular exfoliation of DNA into stool occurs continuously, Cologuard can detect pre-malignant neoplasia at early onset of abnormality." Ex. 1081, 3; Ex. 2032, 1.

The Cologuard collection kit referred to above "includ[es] a stool sample for DNA testing (Container) and a stool sample for Hemoglobin testing (Tube)." Ex. 1082, 15. Patent Owner's Cologuard Patient Guide provides a labeled illustration of this kit, which we reproduce below:



Ex. 2023, 8. The illustration is titled "What is Inside the Cologuard Collection Kit," and shows a shipping box, a sample container, a bracket that appears to be shaped to hold the sample container, a bottle of liquid preservative, a tube, sample labels, and patient instructions. The Patient Guide states that the "bottle of liquid contains a preservative (less than 10% EDTA in Tris buffered solution)," and "[t]he tube contains a 10% albumin in Tris buffered detergent solution with an antimicrobial agent." *Id.*

D. PETITIONER'S ASSERTED GROUNDS FOR UNPATENTABILITY

Petitioner asserts the following grounds for unpatentability:

Ground	Claims Challenged	35 U.S.C. §[2]	Reference(s)/Basis
1	1–9, 11, 14–20	103	Lenhard,[3] Vilkin,[4] Itzkowitz[5]
2	12, 13	103	Lenhard, Vilkin, Itzkowitz, Kanaoka[6]
3	10	103	Lenhard, Vilkin, Itzkowitz, Derks[7]
4	1–9, 11, 14–20	103	Shuber,[8] Vilkin
5	12, 13	103	Shuber, Vilkin, Kanaoka
6	10	103	Shuber, Vilkin, Derks

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[2] The priority date of the '781 patent is February 3, 2009, which is before the relevant AIA revisions to the Patent Act took effect on March 16, 2013. 35 U.S.C. § 100 (note); *see* Ex. 1001, code (22). Therefore, pre-AIA § 103 applies, although the version of the statute is not determinative here.

[3] Lenhard et al., *Analysis of Promoter Methylation in Stool: A Novel Method for the Detection of Colorectal Cancer*, 3 CLIN. GASTROENTEROL. AND HEPATOLOGY 142–49 (2005) (Ex. 1004, "Lenhard").

[4] Vilkin et al., *Performance Characteristics and Evaluation of an Automated-Developed and Quantitative, Immunochemical, Fecal Occult Blood Screening Test*, 100 AM. J. GASTROENTEROL. 2519–25 (2005) (Ex. 1005, "Vilkin").

[5] Itzkowitz et al., *Improved Fecal DNA Test for Colorectal Cancer Screening*, 5 CLIN. GASTROENTEROL. AND HEPATOLOGY 111–17 (2007) (Ex. 1006, "Itzkowitz").

[6] S. Kanaoka, US2006/0216714 A1, pub. Sept. 28, 2006 (Ex. 1007, "Kanaoka").

[7] Derks et al., *Promoter Methylation Precedes Chromosomal Alterations in Colorectal Cancer Development*, 28 CELLULAR ONCOLOGY 247–57 (2006) (Ex. 1008, "Derks").

[8] Shuber et al., WO2005/113769 A1, pub. Dec. 1, 2005 (Ex. 1009, "Shuber").

Pet. 12. In support of these grounds for unpatentability, Petitioner relies on, *inter alia*, the 1st and 2nd Declarations of Duncan Whitney, PhD. Ex. 1002; Ex. 1080. In support of its positions, Patent Owner relies on, *inter alia*, the 1st and 2nd Declarations of Vadim Backman, PhD, and the Declaration of Peter Wood. Ex. 2001; Ex. 2022; Ex. 2023.

II. DISCUSSION

A. LEGAL STANDARDS

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to Patent Owner.[9] *See Dynamic Drinkware*, 800 F.3d at 1378.

The Supreme Court in *KSR International Co. v. Teleflex Inc.*, 550 U.S. 398 (2007), reaffirmed the framework for determining obviousness set forth in *Graham v. John Deere Co.*, 383 U.S. 1 (1966). The *KSR* Court summarized the four factual inquiries set forth in *Graham* (383 U.S. at 17–18) that are applied in determining whether a claim is unpatentable as obvious under 35 U.S.C. § 103 as follows: (1) determining the scope and content of the prior art; (2) ascertaining the differences between the prior art and the claims at issue; (3) resolving the level of ordinary skill in the art; and

[9] Although we may refer to Patent Owner's arguments as "not persuasive," this is in the context of assessing the record and the parties' arguments; we do not shift the ultimate burden from Petitioner.

(4) considering objective evidence indicating obviousness or non-obviousness. *KSR*, 550 U.S. at 406.

"The combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results." *Id.* at 416. "[W]hen the question is whether a patent claiming the combination of elements of prior art is obvious," the answer depends on "whether the improvement is more than the predictable use of prior art elements according to their established functions." *Id.* at 417.

"[O]bjective indicia of non-obviousness play an important role as a guard against the statutorily proscribed hindsight reasoning in the obviousness analysis," and "may often be the most probative and cogent evidence in the record." *WBIP, LLC v. Kohler Co.*, 829 F.3d 1317, 1328 (Fed. Cir. 2016); *see also Graham*, 383 U.S. at 36 ("[Objective indicia] may also serve to guard against slipping into use of hindsight, and to resist the temptation to read into the prior art the teachings of the invention in issue." (internal quotation marks and citation omitted)). "In order to accord substantial weight to [objective indicia of non-obviousness, also called] secondary considerations[,] in an obviousness analysis, the evidence of secondary considerations must have a nexus to the claims, *i.e.*, there must be a legally and factually sufficient connection between the evidence and the patented invention." *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1373 (Fed. Cir. 2019) (citing *Henny Penny Corp. v. Frymaster LLC*, 938 F.3d 1324, 1332 (Fed. Cir. 2019)). "The patentee bears the burden of showing that a nexus exists." *Id.*

B. ORDINARY SKILL IN THE ART

In determining the level of ordinary skill in the art, we may consider, if in evidence, the types of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus. Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986); *Orthopedic Equip. Co. v. United States*, 702 F.2d 1005, 1011 (Fed. Cir. 1983).

Petitioner contends that a person of ordinary skill in the art at the time of the invention (using the acronym "POSA") would have had "a Ph.D. in chemistry, biochemistry, biology, or a related field and at least five years of experience designing and performing diagnostic assays on fecal samples." Pet. 12 (citing Ex. 1002 ¶¶ 9–12).

Patent Owner disagrees.[10] Patent Owner proposes that

> a POSA would have a doctoral degree in medicine, chemistry, biochemistry, biology, or a related field and 1-2 years of experience in the processing and analysis of biological samples,

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[10] In a prior reexamination request over the '781 patent, Petitioner previously argued for the Office to define the ordinarily skilled artisan as having

> at least a Bachelor of Science degree in Biology (or a similar major) and several years of experience in the processing of biological samples, such as fecal samples from a human. Such a person would have been well-acquainted with methods of collecting fecal samples, preserving proteins and/or nucleic acids, including hemoglobin, DNA, and RNA, and subsequently evaluating the fecal samples for proteins and/or nucleic acids, including hemoglobin, DNA, and RNA, as of February 3, 2009.

Resp. 19–20 (quoting Ex. 1021, 7).

> including fecal samples. [Ex. 2022] ¶51. Such a person could have been an individual or a member of a team of scientists addressing fecal sample processing and analysis. *Id.* The Board accepted Patent Owner's definition for purposes of the Institution Decision. [DI] at 8.

Resp 20. Patent Owner argues that Petitioner's proposed definition in this proceeding is overly narrow and Petitioner's proposed definition in the related reexam is too broad, and that Patent Owner's definition falls correctly between the two. *Id.* (citing Ex. 2022 ¶¶ 50–51).

Here, as in our Institution Decision, we do not discern a significant difference between the parties' respective proposed definitions beyond the number of years of experience after obtaining a doctoral degree. *Compare* Pet. 12, *with* Prelim. Resp. 47, *and* Resp. 19–20; *see also* DI 7–8. Under either proposed definition, the ordinarily skilled artisan would have been very highly skilled and educated.

For purposes of this Final Decision, we maintain that Patent Owner's definition appears to be reasonable (and it falls between Petitioner's two definitions asserted in the reexamination request and in the Petition). Further, Patent Owner's proposed definition appears to comport with the level of skill in the art reflected in the prior art of record and the '781 patent itself, for example, Vilkin's authors are a group of medical doctors and an editor of the journal is a PhD, Joost Louwagie, the named inventor of the '781 patent, holds a PhD in Biochemistry, and the authors of Itzkowitz are medical doctors and PhDs. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) ("[T]he prior art itself [may] reflect[] an appropriate level" as evidence of the ordinary level of skill in the art) (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985)); *see* Ex. 1005, 3 (list of authors); Ex. 1016 (Dr. Louwagie profile); Ex. 1055;

see also generally Reply (Petitioner did not contest our similar determination at Institution). Therefore, we adopt and use Patent Owner's definition of the ordinarily skilled artisan, set forth above.

C. CLAIM CONSTRUCTION

The Board interprets claim terms in an *inter partes* review using the same claim construction standard that is used to construe patent claims in a civil action in federal district court. 37 C.F.R. § 42.100(b). In construing claims here, similar to district courts, we by default give claim terms their ordinary and customary meaning, which is "the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention." *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Petitioner asserts that no claim construction is necessary in this proceeding. *See generally* Pet. Similarly, Patent Owner states that it "does not believe that claim construction is necessary to resolve patentability" in this proceeding. Resp. 21.

"[W]e need only construe terms 'that are in controversy, and only to the extent necessary to resolve the controversy.'" *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)). We agree with the parties that no construction of any claim term is necessary for purposes of our Final Decision.

D. PETITIONER'S ASSERTED PRIOR ART

We summarize Petitioner's asserted prior art below. There is no dispute here that Lenhard, Vilkin, Itzkowitz, Kanaoka, Derks, and Shuber are prior art. *See generally* Resp.

1. Lenhard (Ex. 1004)

Lenhard is a journal article titled "Analysis of Promoter Methylation in Stool: A Novel Method for the Detection of Colorectal Cancer," that indicates it was published in the journal *Clinical Gastroenterology and Hepatology* in 2005; it is prior art under 35 U.S.C. § 102(b). Ex. 1004, 142.

According to Lenhard, the detection of tumor-derived genetic changes in stool is a promising approach for CRC screening and the use of fecal occult blood testing (FOBT)[11] alone or in combination with endoscopy, was recommended. *Id.* Lenhard states that, "FOBT, when applied regularly, can decrease CRC [(colorectal cancer)] mortality by only 15–33%." *Id.*

Lenhard describes a study involving the potential use of the gene hypermethylated in cancer 1 ("HIC1") promoter methylation as a stool-based DNA marker. *Id.* at 143. According to Lenhard, the promoter of HIC1 frequently is methylated in CRC, but not in normal or aging colonic tissue. *Id.* Lenhard states that it has shown that "HIC1 promoter methylation can be detected frequently and with high specificity in stool samples from patients with CRCs." *Id.*

[11] As discussed above, and as described in the '781 patent, there are two types of FOBT identified in the record: gFOBT and iFOBT. Ex. 1005, 2519. gFOBT refers to "guaiac fecal occult blood test." *Id.* This test is based on a chemical reaction between heme and guaiac, a compound isolated from Guaiacum trees, causing a color change. Ex. 1002 ¶ 51. iFOBT (sometimes called FIT, fecal immunochemical tests) refers to "immunochemical, fecal occult blood test," which is "specific for human Hb," i.e., human hemoglobin; the test is based on antibody detection of hemoglobin. Ex. 1005, 2519; *see also* Ex. 1002 ¶ 52.

Moreover, Lenhard states, "[t]he combination of HIC1 methylation analysis with FOBT allowed for detection of two thirds of CRCs." *Id.* at 147. According to Lenhard, "[t]he combination of both assays resulted in increased detection rates for CRCs." *Id.*; *see also id.* at 146 (Table 4) (providing data for positivity rates of HIC1 assay, FOBT assay, and combination of the tests). Lenhard further states that "[a]lthough the combined test detected all localized cancers, no increase in sensitivity for adenoma was seen." *Id.*

Lenhard states that stool samples were collected preoperatively from patients with verified CRCs and before colonoscopy for patients with adenomas larger than one centimeter. *Id.* Samples were received within ten hours after defecation at the laboratory, subjected to gFOBT (color change test) immediately on receipt, and then stored at -80°C until analyzed for methylated DNA. *Id.* at 143, 145.

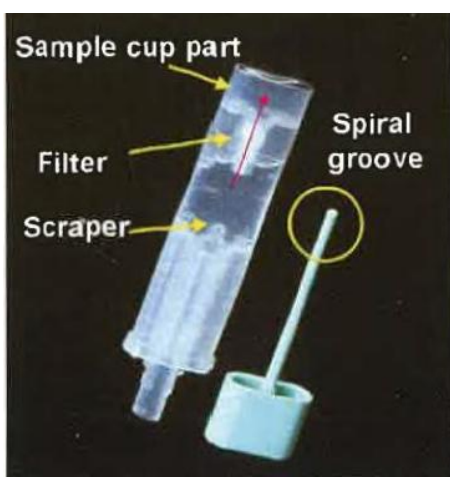
Lenhard expressly suggests performing FOBT in combination with another screening technique, i.e., colonoscopy and DNA testing. *See generally id.* Upon testing by combining FOBT with gene methylation analysis it found the combination effective to detect 2/3 of cancer instances. *Id.* at 146–47. The studied FOBT worked as well as reported in the past and the HIC1 was slightly better (more sensitive, detected some other types of cancer better). *Id.* at 147. Using both tests "resulted in increased detection rates." *Id.* Lenhard compliments the combination of FOBT and HIC1 detection as a "highly sensitive, specific, and easily analyzable" process. *Id.*

2. Vilkin (Ex. 1005)

Vilkin is a journal article titled "Performance Characteristics and Evaluation of an Automated-Developed and Quantitative, Immunochemical, Fecal Occult Blood Screening Test," that indicates it was published in the *American Journal of Gastroenterology* in 2005; it is prior art under 35 U.S.C. § 102(b). Ex. 1005.

Vilkin explains that the standard gFOBT is faulted for its low sensitivity for significant colorectal neoplasia (i.e., CRC and advanced adenomatous polyps ("AAP")), and low specificity due to nonspecificity for human hemoglobin ("Hb"). *Id.* at 2519. Vilkin explains that the introduction of central laboratory and office-developed iFOBT, specific for human Hb, improved specificity. *Id.* Vilkin describes a colonoscopy-controlled study for a detailed evaluation of an automated desktop instrument for quantitative, immunochemical determination of fecal occult blood. *Id.* at 2523.

Vilkin describes a fecal test sampling device shaped like a small test tube with a fecal probe that is inserted into the stool and then pushed back into the tube, through a membrane into a sample cup that includes an Hb stabilizing buffer. *Id.* at 2520. Vilkin's labeled photograph (Figure 1) of this sampling device is reproduced below:



Vilkin's Figure 1, shown in part above, depicts and labels a fecal sample storage tube having a sample cup part, a filter, a scraper, and a cap with a prong and a spiral groove. *Id.* Vilkin states:

> The fecal test sampling device is shaped like a small test tube with the fecal probe inserted into it and sealing it. The probe has a serrated tip, which is poked into the stool and then pushed back into the tube, past a scrapper, and through a membrane into the sample cup. These remove most of the excess feces and leave the stool collected (about 10 mg) into the serrations (Fig. 1). The tip is then put in a closed amount of Hb stabilizing buffer.

Id. Vilkin explains that the samples are double-closed in ziplock bags and kept in the refrigerator until returned to the laboratory where they are kept at 4°C until development. *Id.* Vilkin explains that "[t]he instrument automatically mixes the fecal buffer solution with the latex-antihuman HbA antibody reagent." *Id.*

Vilkin discloses that iFOBT screening provides high sensitivity (76.5%) and acceptable specificity (95.3%; better than gFOBT); identifying six of six CRC patients, twenty of 28 advanced adenomatous polyp patients, and 26 of 34 neoplasia patients. *Id.* at 2522–23. Vilkin suggests a need to preserve the sample with buffer, no need to freeze, and stability for shipping to lab (3 weeks just refrigerated). *Id.* at 2523–24. Vilkin discloses that iFOBT was already proven effective in Japan and Hong Kong, and had the advantages of convenience, cost-reduced screening, and not requiring dietary restrictions. *Id.* at 2519.

3. Itzkowitz (Ex. 1006)

Itzkowitz is a journal article titled "Improved Fecal DNA Test for Colorectal Cancer Screening" that indicates it was published in the journal

Clinical Gastroenterology and Hepatology in 2007; it is prior art under 35 U.S.C. § 102(b). Ex. 1006.

Itzkowitz explains that several studies have shown the feasibility of detecting colon tumor-specific products (i.e., colorectal cancer or CRC) in patient stool. *Id.* at 111. The markers for CRC in these studies represent alterations of genes. *Id.* Itzkowitz teaches that pilot studies showed that several technical and conceptual advances could improve fecal DNA testing sensitivity. *Id.* For example, adding a DNA-stabilizing buffer to the stool immediately after defecation (sample collection) was shown to prevent DNA degradation for several days and enhance the performance of a DNA integrity assay ("DIA"). *Id.* Also, Itzkowitz explains that promoter methylation has become recognized as a key pathway by which colon cancers develop and a marker for CRC. *Id.*

Itzkowitz describes a two-phase study. *Id.* at 112. Phase 1 involved analyzing stool samples from approximately 50 patients with CRC and 200 patients with normal colonoscopy to define suitable DIA cut-off values and to determine optimal markers for the new assay. *Id.* Phase 2, which was then ongoing, was designed as a validation set in which an additional 125 patients with CRC and 200 patients with normal colonoscopy would be analyzed using the optimal marker panel from phase 1. *Id.*

Itzkowitz explains that subjects were given a special stool collection kit to be mounted on a toilet bowl for sample collection. *Id.* Immediately after defecation (in the kit), the subject added 250 ml of a DNA-stabilizing buffer to the stool specimen and then shipped the specimen, at room temperature, overnight, to a laboratory for processing and analyzing. *Id.*

Itzkowitz describes this as “a noninvasive screening test that is convenient, safe, and easy to perform at home.” *Id.* at 115.

Itzkowitz discloses that fecal DNA testing was known to have advantages over existing fecal occult blood tests, that the ability to ship/mail samples for testing removes geographic access barriers, and that if a DNA preservation buffer is used, samples can be preserved, even at room temperature, for shipping to lab. *Id.* at 115. Itkowitz discloses that the observed improved sensitivity relating to the use of a DNA-preserving buffer and the related techniques was at the expense of a minor decrease in specificity, but that the “improved DNA preservation and extraction” technique was still “a better screening test.” *Id.* at 116.

Itzkowitz discloses that, relating to its DNA sampling and preservation technique, “[i]nherent in the success of any screening test is patient acceptance,” and the authors’ experience was “a very high degree of patient satisfaction.” Itzkowitz concludes that “the assay complexity can be reduced significantly, thereby decreasing cost and facilitating the development of an assay kit that could be managed by local clinical laboratories,” and the goal is “to get patients screened who might otherwise avoid an invasive screening test.” *Id.* at 117.

4. Kanaoka (Ex. 1007)

Kanaoka, titled “METHOD OF DETECTING COLON CANCER MARKER,” is the September 28, 2006, publication of U.S. Application 10/549,389, which was filed on September 19, 2003; Kanaoka is prior art under 35 U.S.C. § 102(b). Ex. 1007, codes (10), (43), (21), (22), (45). Kanaoka discloses that its invention relates to the following:

> It is intended to provide a non-invasive and convenient method of detecting a tumor marker for diagnosing colon cancer which

> is superior in sensitivity and specificity to the existing fecal occult blood test. More specifically speaking, a method of detecting a tumor marker for diagnosing colon cancer which comprises collecting biological sample which is immediately frozen using liquid nitrogen in some cases, homogenizing the sample in the presence of an inhibitor of an RNA digesting enzyme to give a suspension, extracting RNA from the obtained suspension, subjecting the extracted RNA to reverse transcription to give cDNA, amplifying the obtained cDNA and then detecting the thus amplified cDNA. This method is characterized by involving no procedure of separating cell components from the biological sample.

Id. at Abstract. Kanaoka states that, "[t]he tumor marker used in the present invention is COX-2, matrixmetalloprotease (MMP), c-met, CD44 variants, EGF-R, EF-1, Wnt-2, Bradeion, SKP2, KPC-1, KPC-2, PRL-3, Angiogenin, Integrin, Snail, Dysadherin, or the like, and is preferably COX-2." *Id.* ¶ 44.

In one example, Kanaoka discloses that feces samples were obtained, separated into 1 g parts in 5 ml tubes, and then frozen. *Id.* ¶ 57. These samples were tested to measure human hemoglobin (Hb) by immunological fecal occult blood testing to obtain a comparison reference value. *Id.* Next, after homogenization, RNA was extracted from the samples and cDNA transcribed, and then amplified by PCR. *Id.* ¶¶ 57–58. Then, using primers designed for COX-2, CEA (carcinoembryonic antigen protein) and COX-2 were detected. Each of CEA, COX-2, and blood via the iFOBT were detected in patients with colon cancer (almost every time). *Id.* ¶¶ 63–66.

5. Derks (Ex. 1008)

Derks, titled "Promoter methylation precedes chromosomal alterations in colorectal cancer development," is an article published in 2006 in the journal *Cellular Oncology*; Derks is prior art under 35 U.S.C. § 102(b). Ex. 1008, 247.

Derks states that "[c]olorectal cancers are characterized by genetic and epigenetic alterations" and its reported "study aimed to explore the timing of promoter methylation and relationship with mutations and chromosomal alterations in colorectal carcinogenesis." *Id.* Derks states its study "evaluated promoter methylation status of *hMLH1, O*^{6} *MGMT, APC, p14*^{ARF}*, p16*^{INK4A}*, RASSF1A, GATA-4, GATA-5,* and *CHFR* using methylation-specific PCR. Mutation status of *TP53, APC* and *KRAS* genes were studied by p53 immunohistochemistry and sequencing of the *APC* and *KRAS* mutation cluster regions." *Id.* Derks "conclude[d] that promoter methylation can be regarded as an early event preceding TP53 mutation and chromosomal abnormalities in colorectal cancer development." *Id.*

6. Shuber (Ex. 1009)

Shuber, titled "Method for Stabilizing Biological Samples for Nucleic Acid Analysis," is the December 1, 2005 published version of PCT Application PCT/US2005/017046; Shuber is prior art under 35 U.S.C. § 102(b). Ex. 1009, codes (12), (21), (43), (54).

Shuber states, "[d]isclosed is a method for preparing nucleic acid containing biological samples for a nucleic acid integrity assay and/or multiple mutation analysis by incubating the biological sample with a stabilization solution that includes a buffer, a chelating agent and a salt." *Id.* at Abstract.

Shuber discloses methods for preparing nucleic acid-containing biological samples for an assay to detect nucleic acid markers indicative of cancer. *Id.* at 1:11–13. According to Shuber, contacting a patient sample with a stabilization solution stabilizes the DNA so that intact nucleic acids indicative of diseased cells are more effectively detected in a nucleic acid integrity assay. *Id.* at 2:7–9. Shuber explains that a stabilization solution "may be particularly useful when samples are not refrigerated or frozen" and if a sample "is obtained at a remote location and mailed or delivered to a testing center." *Id.* at 10:10–14.

In one aspect of the Shuber invention, a stool sample may be directly deposited into a sealable container and a stabilization solution may be added to the container, after which the container may be sealed for storage/shipping to a test lab. *Id.* at 29:8–19.

E. OBJECTIVE EVIDENCE OF NON-OBVIOUSNESS

This decision ultimately analyzes all *Graham* factors, but, because each of the Grounds for unpatentability presented by Petitioner invokes the same objective indicia considerations, we address those considerations at the outset. In opposition to each of Petitioner's grounds for unpatentability, Patent Owner argues that objective indicia of non-obviousness (which Patent Owner refers to as "secondary considerations"), in the form of commercial success, failures of others, long-felt and unmet need, industry praise, and copying, provide evidence of the claims' patentability. Resp. 50–66. Petitioner argues, *inter alia*, that Patent Owner's position on objective indicia of non-obviousness fails from the outset because Patent Owner does not establish a nexus exists between the evidence and the challenged claims. Reply 22–27.

1. Nexus

As noted above, a nexus is critical to objective indicia of non-obviousness, and the Federal Circuit instructs that "[t]he patentee bears the burden of showing that a nexus exists." *Fox Factory*, 944 F.3d at 1373 (quoting *WMS Gaming Inc. v. Int'l Game Tech.*, 184 F.3d 1339, 1359 (Fed.

Cir. 1999)). And, the Federal Circuit further instructs that, "[t]o determine whether the patentee has met that burden, we consider the correspondence between the objective evidence and the claim scope." *Id.* (quoting *Henny Penny*, 938 F.3d at 1332).

A rebuttable presumption of nexus between the asserted evidence of objective indicia of non-obviousness and a patent claim is established if the patent owner shows that evidence is tied to a specific product, which "*is the invention* disclosed and claimed," i.e., the product "'embodies the claimed features, and is coextensive with them.'" *Demaco Corp. v. F. Von Langsdorff Licensing Ltd.*, 851 F.2d 1387, 1392 (Fed. Cir. 1988) (emphasis added); *Polaris Indus., Inc. v. Arctic Cat, Inc.*, 882 F.3d 1056, 1072 (Fed. Cir. 2018) (quoting *Brown & Williamson Tobacco Corp. v. Philip Morris Inc.*, 229 F.3d 1120, 1130 (Fed. Cir. 2000)). "Although we do not require the patentee to prove perfect correspondence to meet the coextensiveness requirement, what we do require is that the patentee demonstrate that *the product is essentially the claimed invention*." *Fox Factory*, 944 F.3d at 1374 (emphasis added).

Patent Owner points to the Cologuard product for the nexus required of its objective indicia of non-obviousness evidence. Resp. 50. More specifically, Patent Owner walks through the disclosure of the Cologuard Patient Guide (Ex. 2032) and maps it to the limitations of claims 1 and 3 to establish that the product embodies the claims and is coextensive therewith. *Id.* at 51–55 (citing Ex. 2022 ¶¶ 143–154; Ex. 2032, 6, 8, 14, 17–18, 20–28, 30–31, 46). Patent Owner's witness, Dr. Backman, testifies that there are "two critical aspects of screening tests," such as the Cologuard product's screening test, that result in product adoption: "diagnostic accuracy and uptake," that is, "diagnostic performance" and "convenien[ce]." Ex. 1022 ¶ 55.

For example, regarding commercial success, Patent Owner asserts Cologuard sales have grown over the years to the millions of tests and hundreds of millions (or, now, billions) of dollars in revenue. Resp. 55–58 (citing, e.g., Ex. 2023 ¶¶ 36–38; Ex. 2029, 11; Ex. 2036; Ex. 2038 ¶ 40). Regarding the failures of others and the satisfaction of a long-felt and previously unmet need, Patent Owner points to CRC screening testing available (or not) before Cologuard, the introduction of Cologuard, and its success. *Id.* at 58–62. As for evidence of industry praise, again Patent Owner identifies that Cologuard has repeatedly been the subject of awards and praise. *Id.* at 62–63. Finally, regarding copying, Patent Owner asserts that Petitioner's "ColoSense Collection Kit contains the same elements as Exact Science' successful Cologuard Collection Kit." *Id.* at 63–66.

We have no reasonable doubt that *aspects* of the Cologuard product practice the invention of the '781 patent's claims 1 and 3, which are most generally directed to steps of collecting two portions of a fecal sample at home, and combining a respective buffer stabilizer/preservative with each of the two portions for claim 1, and then processing those fecal samples by testing for blood and nucleic acids (e.g., DNA) for claim 3. These appear to be nearly the most basic steps for using the Cologuard kit. *See generally* Ex. 2032. And, these steps do illustrate a certain ease of use and consumer-friendliness that may be a contributing factor in Cologuard's success. *See, e.g.*, Ex. 2022 ¶¶ 140–142, 155, 158–159, 170–171.

However, Petitioner argues that a nexus cannot be presumed to exist between the claims and the Cologuard product and, furthermore, that Patent

Owner has not otherwise demonstrated a nexus, because Cologuard is "clearly not coextensive with the claimed method." Reply 22 (citing Ex. 1080 ¶¶ 99–106). Petitioner argues that Cologuard is a multi-faceted product of which the components and processes covered by the '781 patent are but a part, which is decisively illustrated by the fact that Patent Owner has marked the Cologuard product with dozens of other patents. *Id.* at 22–23 (citing Ex. 1080 ¶ 105; Ex. 1081, 4; Ex. 1082, 7–46; Ex. 1083; Ex. 1084); *see infra* at 28–29 (listing such patents). Petitioner also identifies that Patent Owner itself has expressly stated that Cologuard's "diagnostic accuracy 'may be attributed to the DNA marker and algorithm components of the test,'" and has also expressly attested to the Office that Cologuard's success is attributable to a (later-patented, subsequent to the '781 patent) "combination of markers and methods for extracting the marker DNA." *Id.* at 24 (citing Ex. 1080 ¶¶ 111–112; Ex. 1085; Ex. 1086, 6–7, 9 (arguing to a patent examiner in an "After Final Amendment" dated July 31, 2015 in U.S. application 14/145,082, which issued as US 9,163,278 B2 ("the '278 patent")[12] on Oct. 20, 2015, that these later-claimed inventions were the reasons for Cologuard's success and satisfaction of the long-felt need); Ex. 2008, 1295).

We agree with Petitioner and are persuaded by its arguments.

The Federal Circuit addressed facts similar to those before us here in *Fox Factory*, which dealt extensively with the issue of nexus and objective indicia of non-obviousness (which the Federal Circuit there called

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[12] Office records show the '278 patent includes claims to "[a] system for isolating a target human DNA from a human stool sample."

"secondary considerations"). *Fox Factory*, 944 F.3d at 1372. At the outset, the Court identified that "[i]n order to accord substantial weight to secondary considerations in an obviousness analysis, 'the evidence of secondary considerations must have a 'nexus' to the claims, i.e., there must be 'a legally and factually sufficient connection' between the evidence and the patented invention.'" *Id.* at 1373 (quoting *Henny Penny*, 938 F.3d at 1332). The *Fox Factory* decision also addressed coextensiveness, a question of fact, stating that it is the cornerstone of the nexus requirement, that it is rarely shown by "perfect correspondence between the claimed invention and the product," but that nexus is not present where "the patented invention is only a component of a commercially successful machine or process." *Id.* at 1373–74.

Fox Factory involved *two* patents, each with claims directed to aspects of bicycle chainrings (i.e., the gears turned by a bike's pedals) and the configuration of teeth thereon, and a product called X-sync asserted to be the embodiment of the invention(s). *Id.* at 1369–70. The record of *Fox Factory* evidenced that the X-sync product indisputably included features that were "critical" and "more than an insignificant feature," but not claimed in the first patent. *Id.* at 1374–75. These features of X-sync, unclaimed in that first patent, were touted by patentee as addressed by the second patent. *Id.* The patentee, in two separate proceedings before the Board over these two separate patents, had held up the X-sync product as evidencing objective indicia of non-obviousness for separate features of the product, which were claimed separately in the two different patents.

The Federal Circuit held that there was no nexus on such evidence (although the patentee would be given a chance to prove nexus on remand).

Id. at 1378. In so deciding, the Federal Circuit held that when there are features not claimed in a patent that materially impact the functionality of the product, nexus is not presumed. *Id.* at 1376. In later decisions, the Federal Circuit has clarified that *critical* unclaimed features are of particular importance. *Teva Pharmas. Int'l GmbH v. Eli Lilly and Co.*, 8 F.4th 1349, 1362–63 (Fed. Cir. 2021). Moreover, when analyzing the coextensiveness required for nexus, it cannot be reduced to "an inquiry into whether the patent claims broadly cover the product;" the product must *be* "the invention disclosed and claimed," and *not* "only a component of a commercially successful machine or process." *Fox Factory*, 944 F.3d at 1377 (quoting *Demaco*, 851 F.2d at 1392).

Fox Factory involved evidence on only *two* patents covering a single product. Here, there are *dozens* of patents, in addition to the '781 patent, identified as covering Cologuard. *See* Ex. 1084 (Exact Sciences' website marking Cologuard patents). Cited Exhibit 1084 states:

> The Cologuard® test and its use are covered by one or more of the following patents, associated pending patent applications, or non-U.S. international equivalents: U.S. 8,361,720, 8,673,555, 8,715,937, 8,722,330, 8,916,344, 8,969,046, 8,980,107, 8,993,341, 8,999,176, 9,000,146, 9,024,006, 9,057,098, 9,121,070, 9,121,071, 9,127,318, ***9,163,278***, 9,169,511, 9,211,112, 9,212,392, 9,290,797, 9,315,853, 9,376,721, 9,399,800, 9,631,228, 9,637,792, 9,657,330, 9,750,482, 9,783,856, 9,803,249, 9,845,491, 9,856,515, 9,982,308, 10,000,817, 10,047,390, 10,093,987, 10,106,844, 10,138,524, 10,144,953, 10,196,676, 10,253,358, 10,265,054, D849,266, 10,435,753, 10,450,614, 10,486,158, 10,519,510 10,590,489, 10,604,793, 10,626,465, 10,702,250, 10,704,083, 10,808,286, 10,822,639, 10,870,893, 10,889,867, 10,894,256, ***11,634,781***, 11,845,991, and 11,970,746. This product and its use are also covered by patents pending in the U.S. and around the world.

Ex. 1084, 1 (emphasis added to highlight the ’781 patent challenged here and the ’278 patent, identified by Patent Owner as claiming particularly important elements of the Cologuard Product).

Furthermore, during prosecution of the ’278 patent, Patent Owner expressed that elements of Cologuard not covered by the ’781 patent, but covered by the ’278 patent (i.e., “a particular combination of markers and methods for extracting the maker DNA,” “particular combination of ‘genes of interest,’” the “particular combination of reagents,” and also, potentially (there is no direct evidence this is covered by another patent), the diagnostic accuracy provided by “the DNA marker and algorithm components of the test”) are reasons for Cologuard’s successes where others failed.[13] *See* Ex. 1086, 6–9 (note, this exhibit at page 7 cites the following Exhibit 2008 as evidence supporting Patent Owner’s prosecution arguments); Ex. 2008, 1294–95; *see also* Hr’g Tr. 44:12–45:4 (Patent Owner argued at the Final Hearing here that “detection of adenomas . . . is a critical feature”—in our reading, this is not expressly recited in or required by any claim of the ’781 patent).

Thus, even though aspects of the Cologuard product appear to embody claims of the ’781 patent, the product has a multitude of features not

[13] Patent Owner expressed that these unclaimed (by the ’781 patent) features “addressed [the] long-felt need and succeeded where others had failed, as evidenced by the first-ever FDA approval for the product containing these components.” Ex. 1086, 9. The fact that other patented inventions (of the ’278 patent), not at issue in this proceeding, met the long-felt need and were the reason for success (of Cologuard) supports our giving little weight to Patent Owner’s arguments on long-felt need and failures of others/success of Patent Owner.

covered by the ’781 patent’s claims, some of which are critical unclaimed elements according to Patent Owner. Accordingly, Cologuard is not coextensive with the challenged claims, and no nexus between the product and the challenged claims is shown on this record.

Other than mapping claims 1 and 3 to aspects of Cologuard, which we explained above is not indicative of the complete product, and asserting Cologuard embodies such claimed features to establish a nexus, Patent Owner does not address its many other patents that purportedly cover Cologuard, or its prior statements to the Office regarding important elements of Cologuard covered by the claims of other patents. Resp. 50–55; Sur-reply 18–20.

Although there is no nexus established here, for completeness we address Patent Owner’s specific asserted objective indicia arguments below. Patent Owner asserts commercial success, satisfaction of a long-felt need, praise, and copying (Resp. 55–66), and we discuss below the flaws in Patent Owner’s positions, including why Patent Owner has not persuasively established coextensiveness with the claims of the ’781 patent sufficient to establish a nexus for these indicia.

2. Commercial Success, Long-Felt Need (and Failures), Praise, and Copying

Where a party is not entitled to a presumption of nexus, it bears the burden of showing that the success of the product is “a direct result of unique characteristics of the claimed invention.” *Fox Factory, Inc.*, 944 F.3d at 1373–74. Patent Owner does not show here that success is the “direct result of unique characteristics of the claimed invention.” For each of the alleged objective indicia, we address whether the evidence supports the existence of a nexus even absent a presumption.

Regarding commercial success, Patent Owner asserts that the billions in revenue from sales of Cologuard (through 2023), due to the "diagnostic accuracy and uptake" of the product, are reflected in the '781 patent's claims, particularly the recited addition of buffers to the sample portions and the recited "at-home collection," and are evidence of the non-obviousness of the challenged claims. Resp. 55–58; *see also* Ex. 2023 ¶ 37. We agree that Cologuard sales have been substantial. *See, e.g.*, Ex. 2038, 40. However, the record does not support that Cologuard's sales are due to the invention claimed in the '781 patent. Petitioner points out that the Cologuard product had and has market exclusivity (provided by, e.g., other patents and FDA approval), and that Patent Owner has publicly stated that it spent over $3 billion on sales/marketing for Cologuard. Reply 26 (citing Ex. 1087; Ex. 2023 ¶ 13); *see also* Ex. 1086, 9 (identifying Cologuard as the "first-ever FDA approval for the product containing . . . components" claimed by another patent). "[T]here may be many reasons a product is commercially successful; it is only where the success is due to the claimed invention that commercial success can show nonobviousness," and here there is evidence that factors other than the '781 patent's claimed invention may have contributed to Cologuard's commercial success. *UCB, Inc. v. Actavis Labs. UT, Inc.*, 65 F.4th 679, 695 (Fed. Cir. 2023). Patent Owner has not established that Cologuard's commercial success is due to the '781 patent's claimed invention, rather than other factors and, for this reason as well, we give this evidence little weight.

Regarding the asserted long-felt need, Patent Owner argues that the accessibility of the Cologuard product for patients satisfied a need for "a CRC screening test with ***sufficient diagnostic accuracy***," and that "low screening levels of individuals within the recommended screening groups persisted until the introduction of Cologuard." Resp. 60–61 (emphasis added) (citing Ex. 2022 ¶¶ 169–170, 172–174; Ex. 2049, 2; Ex. 2050, 10; Ex. 2052, 4–5). Patent Owner argues that Cologuard met these needs "due to [its] non-invasive nature and lack of complications," which addressed "all major limitations of the other attempts at CRC screening." *Id.* The Cologuard product appears to meet the above needs, and has been accepted and purchased by a large population of patients, as asserted by Patent Owner, but there is no persuasive evidence that this is due to the invention claimed in the '781 patent rather than the inventions covered by the dozens of other Cologuard-protecting patents, including the ***diagnostic accuracy*** attributed to the DNA marker and algorithm components, and methods for extracting DNA, claimed ***in the '278 patent***, which appear to at least contribute to meeting the alleged long-felt need. *See* Ex. 1086, 6–9; Ex. 1085; Reply 24–25; *see also* Ex. 2008, 1295 (diagnostic accuracy attributable to DNA markers and algorithm of test). For these reasons also, we give this evidence little weight.

As for the asserted industry praise, Patent Owner identifies several instances where Cologuard has won awards or earned public praise, for example, as the best medical technology product of 2016, the "best of what's new" in 2014, and as an "emerging tech" in 2015. Resp. 62–63. But, other than recognizing the product *generally* as innovative and praiseworthy, it is not apparent from the record that such praise can be attributed to *the invention* of the challenged claims rather than some other element(s) of Cologuard covered by its dozens of other patents, as discussed above. *See*

Ex. 2042; Ex. 2043; Ex. 2044; Ex. 2045; Ex. 2046; *see also* Ex. 1086, 6–9; Ex. 2008. For this reason, also, we give this evidence little weight.

Finally, Patent Owner has identified that Petitioner has developed a product called ColoSense that Patent Owner alleges copies "the same elements as Exact Sciences' successful Cologuard Collection Kit." Resp. 63. At a high level, the ColoSense kit appears similar to Cologuard because it appears to be intended for providing a fecal sample at home for CRC screening (by detecting nucleic acids and blood) and its kit has containers for two portioned parts of the sample with respective preservatives/buffers. *Id.* at 64–65 (citing Ex. 2053, 37–40). Patent Owner also, however, identifies in this same argument that "[n]ucleic acid is extracted" from one portion of the sample and "tested for an amount of nucleic acid," which is subject matter relating to claims of other patents covering Cologuard (or at least not covered by the '781 patent). *Id.* at 65 (citing Ex. 2053, 54–56); *see also* Ex. 1086, 6–9; Ex. 2008, 1295. It is unclear whether the asserted copying is directed to the challenged claims or the claims of other Cologuard patents, or some combination thereof. Because Patent Owner's Cologuard cannot be said to be coextensive with the challenged claims, evidence of copying Cologuard, even if taken as true, cannot be given substantial weight because of the same nexus issues discussed above. Moreover, "copying 'requires evidence of efforts to replicate a specific product,'" and even *established* "infringement, taken alone, is not probative of copying." *Tokai Corp. v. Easton Ent., Inc.*, 632 F.3d 1358, 1370 (Fed. Cir. 2011) (quoting *Wyers v. Master Lock Co.*, 616 F.3d 1231, 1246 (Fed. Cir. 2010); *Iron Grip Barbell Co., Inc. v. USA Sports, Inc.*, 392 F.3d 1317, 1325 (Fed. Cir. 2004)). Similarity between products, which is the only evidence of record with respect to copying, does not establish copying. For these reasons, we give this evidence little weight.

3. Summary and Conclusion

For the reasons discussed above, and because the nexus issue affects each basis for objective indicia of non-obvious asserted by Patent Owner, we give no substantial weight to any of Patent Owner's arguments on such indicia. *In re Affinity Labs of Texas, LLC*, 856 F.3d 883, 901 (Fed. Cir. 2017) (without an established nexus between the claimed invention and asserted objective indicia of non-obviousness, such arguments are accorded little weight).

In any event, even if any of the alleged objective indica of non-obvious were entitled to some weight, after analysis of all the obviousness considerations on the complete record as discussed below, we find that the evidence of obviousness on this record is quite strong and substantially outweighs any indicia of non-obviousness on this unique record. *Pfizer, Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1372 (Fed. Cir. 2007) ("Although secondary considerations must be taken into account, they do not necessarily control the obviousness conclusion.").

F. OBVIOUSNESS OVER LENHARD, VILKIN, AND ITZKOWITZ, ALSO COMBINED WITH KANAOKA OR DERKS (GROUNDS 1, 2, AND 3)

Under Ground 1, Petitioner asserts that the combination of Lenhard, Vilkin, and Itzkowitz would have rendered claims 1–9, 11, and 14–20 obvious and, therefore, these claims are unpatentable. Pet. 12, 20–43 (citing Ex. 1001, 12:35–39, 12:47–49; Ex. 1002 ¶¶ 142–161, 163–223; Ex. 1004, 142–145, 147 (and generally); Ex. 1005, 2519–24, Fig. 1, Table 1 (and generally); Ex. 1006, 111–12, 115–17 (and generally); Ex. 1010, 210, 214; Ex. 1024, 29–30; Ex. 1025, 186, 190; Ex. 1028, 7:45–46, Fig. 2; Ex. 1029, 140). Under Ground 2, Petitioner addresses dependent claims 12 and 13 and asserts that this same prior art combination, with the addition of Kanaoka, would have rendered the claims obvious and unpatentable. Pet. 12, 43–46 (citing Ex. 1002 ¶¶ 224–230; Ex. 1007 ¶¶ 21–26, 47–48, 55–68, 76; Ex. 1030, 131). Under Ground 3, Petitioner asserts that dependent claim 10 would have been obvious and unpatentable over the same prior art combination as in Ground 1, with the addition of Derks. Pet. 12, 46–47 (citing Ex. 1002 ¶¶ 231–234; Ex. 1004, 148; Ex. 1008, 252, Table 3b).

Patent Owner opposes each of these grounds for unpatentability. Resp. 21–39 (citing Ex. 1002 ¶ 74; Ex. 1004, 2–3, 5–7, Tables 2, 4 (and generally); Ex. 1005, 3–4, 6–8, Table 3; Ex. 1006, 2, 6–7; Ex. 1013, 7; Ex. 1024, 4, 29; Ex. 2022 ¶¶ 29, 56, 58, 61–62, 67, 69–70, 76, 95–119; Ex. 2027, 148:14–22, 149:18–150:20, 169:13–17); *see also* Sur-reply. Patent Owner argues Grounds 1–3 together as a group, indicating that Grounds 2 and 3 fall (or, conversely, stand) with Ground 1, and also asserting that Kanaoka (relating to Ground 2), like Lenhard, would not have motivated the ordinarily skilled artisan to combine the prior art to achieve the invention, but would have discouraged them. *Id.* at 49 (citing Ex. 1007 ¶¶ 65–67; Ex. 2022 ¶ 138).

We summarize below the parties' positions and provide our reasoning and analysis supporting our conclusions.

1. Petitioner's Positions

Petitioner asserts that: (1) Lenhard discloses a non-invasive CRC screening method where a fecal sample is separated into two portions so that a first portion is tested for blood using gFOBT and the second portion is tested for tumor-derived DNA (i.e., nucleic acids) using methylation of the

HIC1 gene promoter as a detectable biomarker; (2) Vilkin discloses an improved fecal-blood-detection technique, that is, iFOBT paired with buffer stabilization of the sample (suggested to be collected at home, and separated from a larger sample), which would have been an obvious replacement for and improvement on Lenhard's gFOBT technique; and (3) Itzkowitz discloses that adding a DNA stabilizing buffer to a fecal sample to be used for CRC screening (collected at home) improves the technique and would be an obvious addition to Lenhard's process. Pet. 20–26 (citing Ex. 1002 ¶¶ 142–161; Ex. 1004, 142–145, 147; Ex. 1005, 2519–20, 2524; Ex. 1006, 111–12, 116–17; Ex. 1010, 214; Ex. 1024, 29–30; Ex. 1025, 186, 190).

Petitioner asserts that the ordinarily skilled artisan would have recognized from Lenhard that combining genetic analysis and blood detection in respectively portioned stool samples was advantageous because it allowed for the detection of "two thirds of CRCs," increasing the detection rate over either technique alone. *Id.* at 21 (citing Ex. 1002 ¶ 142; Ex. 1004, 142–45); *see also* Ex. 1002 ¶ 140 ("A POSA would understand from the disclosure of Lenhard that combining a DNA methylation test with FOBT resulted in a diagnostic process that is more sensitive than when either test was performed alone.").

Moreover, Petitioner asserts that the ordinarily skilled artisan would have recognized from Vilkin (and the relevant prior art, generally) that Lenhard's gFOBT technique was inferior to Vilkin's iFOBT+stabilizing buffer technique for fecal blood detection because the latter used a known automated analysis, had higher sensitivity and acceptable specificity, did not require dietary restrictions, used a preferred sample handling method, was quantitative so as to allow the physician to set optimal fecal Hb thresholds for a patient, and would permit at-home sample collection. Pet. 21–23 (citing Ex. 1002 ¶¶ 143–147; Ex. 1005, 2519–20, 2524; Ex. 1010, 214; Ex. 1024, 29–30).

Finally, Petitioner asserts that the ordinarily skilled artisan would have recognized from Itzkowitz that collecting a fecal sample directly in a sealable container (as was already known) and using a DNA stabilizing buffer (also already well-known) would have been obvious improvements to Lenhard's two-test screening method because the sample could be reliably shipped to a diagnostic lab for analysis without freezing, and patient satisfaction would be improved. *Id.* at 23–26 (citing Ex. 1002 ¶¶ 148–158; Ex. 1006, 111–12, 116–17; Ex. 1025, 186, 190).

Petitioner asserts that the combination of Lenhard, Vilkin, and Itzkowitz would improve the screening technique introduced by Lenhard for the reasons above and, so, the person of ordinary skill in the art would have been motivated to make the prior art combination and, because combining the features of these three references would have been routine, the ordinarily skilled artisan would have had a reasonable expectation of success. *Id.* at 26 (citing Ex. 1002 ¶ 161).

After addressing the reasons for combining the asserted prior art, Petitioner maps the disclosures of the references to the limitations and elements of the challenged claims.

Regarding independent claim 1, if its preamble ("A method of processing a freshly-collected fecal sample without freezing") is a limitation, Petitioner asserts it is taught by Lenhard's disclosure of sample collection and portioning, Vilkin's disclosure of iFOBT+stabilizing buffer and Itzkowitz's disclosure of DNA stabilizing buffer to be used in Lenhard's method, where the use of stabilizing buffers mean the sample needs at most refrigeration and not freezing. *Id.* at 26–27 (citing Ex. 1002 ¶¶ 164–167; Ex. 1005, 2520; Ex. 1006, 112).

Moving on to the first clause (a) of claim 1 ("collecting a fecal sample from a human subject, wherein the fecal sample is collected at home by the human subject by defecation directly into a sealable collection vessel"), Petitioner asserts this is taught by Itkowitz's disclosure of patients collecting fecal samples directly in a sealable container, on a toilet, at home, which sample is then shipped to a lab. *Id.* at 27–28 (citing Ex. 1002 ¶¶ 169–172; Ex. 1006, 112, 115; Ex. 1026, 2; Ex. 1028, 7:45–49, Fig. 2).

Regarding the next clause (b) of independent claim 1 ("removing a portion of the fecal sample to a separate sealable container to produce a removed portion and a remaining portion of the fecal sample"), Petitioner asserts Lenhard teaches that a fecal sample is divided in two portions, one to be analyzed for blood and one to be analyzed for DNA, each in a separate sealable container as taught by Vilkin. *Id.* at 28–30 (citing Ex. 1002 ¶¶ 173–176; Ex. 1004, 143–45; Ex. 1005, 2520, Fig. 1).

As to the penultimate clause (c) of claim 1 ("combining the removed portion of the fecal sample in the separate sealable container with a buffer that prevents denaturation or degradation of blood proteins found in a fecal sample, and sealing the sealable container"), Petitioner asserts that Vilkin teaches this in disclosing its fecal sample is added to a sample storage tube with Hb stabilizing buffer, which is closed and sealed. *Id.* at 30–31 (citing Ex. 1002 ¶¶ 177–178; Ex. 1005, 2520–21).

As to claim 1's final clause (d) ("combining the remaining portion of the fecal sample in the sealable collection vessel with a stabilizing buffer, and sealing the collection vessel"), Petitioner asserts it is taught by Itzkowitz's disclosure of providing DNA stabilizing buffer with its fecal sample to prevent DNA degradation, in a collection vessel, that is sealed. *Id.* at 31–32 (citing Ex. 1002 ¶¶ 163–183; Ex. 1006, 111–12, 116–17).

Petitioner also maps the disclosure of the prior art to each dependent claim of the '781 patent.

Regarding claim 2, which depends from claim 1 and further requires "delivering [the portioned sample] to a medical diagnostics laboratory," Petitioner asserts the prior art combination teaches this because each reference discloses delivering fecal samples to a laboratory for testing. Pet. 32–33 (citing Ex. 1002 ¶¶ 184–186; Ex. 1004, 143; Ex. 1005, 2520; Ex. 1006, 112). This factual assertion is not specifically disputed by Patent Owner. *See* Resp. 21–39, 49; Sur-reply 2–14.

Claim 3 is directed to a method of processing the portioned sample of claim 1 by testing the first sample portion "for an amount of blood protein" and testing the second sample portion, after "extracting nucleic acid," for "an amount of human nucleic acid," which Petitioner asserts is taught by Lenhard's and Vilkin's disclosing FOBT testing of fecal samples, and Lenhard's and Itzkowitz's disclosing isolation of DNA and DNA testing of fecal samples. Pet. 33–35 (citing Ex. 1002 ¶¶ 189–198; Ex. 1004, 143–44; Ex. 1005, 2519, 2522, Table 1; Ex. 1006, 112). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 21–39, 49; Sur-reply 2–14.

Regarding claim 4, which requires "determining expression from a human gene" when testing for the nucleic acid of claim 3, claim 5, which requires that such expression comprises identifying "an epigenic modification," and claim 6, which requires "measuring an amount of a methylated human DNA," Petitioner asserts that Lenhard and Itzkowitz each teaches this claimed subject matter in disclosing testing for methylated human DNA in fecal samples. Pet. 36–38 (citing Ex. 1001, 12:35–39 (describing "epigenetic modification"); Ex. 1002 ¶¶ 194, 199–206; Ex. 1004, 142, 144; Ex. 1006, 111). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 21–39, 49; Sur-reply 2–14.

Regarding claims 7, 8, and 9, which respectively require that the epigenetic modification analyzed "comprises aberrant methylation," that the "aberrant methylation comprises hypermethylation," and that the human DNA "comprises a gene and/or a promoter region of a gene," Petitioner asserts this is taught by Lenhard disclosing testing fecal samples for hypermethylated HIC1 promoter DNA and by Itzkowitz disclosing testing for vimentin gene hypermethylation. Pet. 38–39 (citing Ex. 1001, 12:47–49 (describing "aberrant methylation"); Ex. 1002 ¶¶ 207–212; Ex. 1004, 142–43; Ex. 1006, 115). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 21–39, 49; Sur-reply 2–14.

Regarding claim 11, which requires "modifying the nucleic acid with bisulfate ions under conditions where[] unmethylated cytosine is converted to uracil," Petitioner again points to Lenhard's disclosure of detecting HIC1 promoter methylation and Itzkowitz's disclosure of testing for vimentin gene methylation, each of which, according to Petitioner, detects DNA methylation using bisulfite conversion, where bisulfite ions are used to convert unmethylated cytosine to uracil (asserting the claim's reference to "bisulfate" instead of "bisulfite" is a typographical error). Pet. 39–40 (citing

Ex. 1002 ¶¶ 105 (testifying that no assay for epigenetic modification modifies nucleic acid with "bisulfate" and the '781 patent describes "bisulfite" at Ex. 1001, 13:34–63, 36:52–57, 37:1–2), 213–215; Ex. 1004, 144; Ex. 1006, 112). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 21–39, 49; Sur-reply 2–14.

Claim 14 requires that, in the method of claim 3, testing for blood protein "comprises testing for a concentration of hemoglobin," and claim 15 requires that this testing for hemoglobin "comprises immunochemical detection of hemoglobin," which Petitioner asserts is taught by Vilkin in disclosing iFOBT, which (quantitatively) tests for hemoglobin. Pet. 40–41 (citing Ex. 1002 ¶¶ 216–219; Ex. 1005, 2519–22, Table 1). This factual assertion is not specifically disputed by Patent Owner. *See* Resp. 21–39, 49; Sur-reply 2–14.

Regarding claims 16–20, each requires detecting a certain threshold of hemoglobin in the fecal sample, i.e., at least 5, 10, 20, 50, or 200 ng/ml, Petitioner asserts such would have been obvious over the prior art combination because these thresholds amount to routine optimization of a result-effective variable; Vilkin explains that the physician can choose an optimal fecal hemoglobin threshold to trigger a follow-up colonoscopy. Pet. 41–43 (citing Ex. 1002 ¶¶ 221–223; Ex. 1005, 2520–21, 2523–24; Ex. 1010, 210; Ex. 1029, 140). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 21–39, 49; Sur-reply 2–14.

As noted above, regarding claims 12 and 13, Petitioner adds the teachings of Kanaoka to the Lenhard-Vilkin-Itzkowitz combination in asserting obviousness. Claim 12 requires measuring RNA expressed from a gene and claim 13 requires that the RNA is measured using reverse transcriptase polymerase chain reaction (RT-PCR), each of which Petitioner asserts is taught by Kanaoka in disclosing a COX-2 RNA biomarker detection process, using RT-PCR, which would be combined with the other prior art and used with a reasonable expectation of success because Kanaoka's successful working example "has a higher detection sensitivity," and would complement the other references' detection processes. Pet. 43–45 (citing Ex. 1002 ¶¶ 224–230; Ex. 1007 ¶¶ 21–26, 47–48, 55–68, 76; Ex. 1030, 131).

As noted above, regarding claim 10, Petitioner adds the teachings of Derks to the Lenhard-Vilkin-Itzkowitz combination in asserting obviousness. Claim 10 requires that the gene having an epigenetic modification be selected from the group consisting of PHACTR3, NDRG4, FOXE1, GATA4, GPNMB, TFP12, SOX17, SYNE1, LAMA, MMP2, OSMR, SFRP2, and CDO1, and Petitioner asserts this is taught by Derks in disclosing testing for GATA-4, which Petitioner asserts would be combined with Lenhard's method, with a reasonable expectation of success, as another biomarker because Lenhard suggests adding more markers for higher sensitivity and specificity. *Id.* at 46–47 (citing Ex. 1002 ¶¶ 231–234; Ex. 1004, 148; Ex. 1008, 250, 252, Table 3b). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

Having summarized Petitioner's positions on the prior art combinations and their teaching of the claimed subject matter, we turn to Patent Owner's positions in opposition.

2. Patent Owner's Positions

As noted above, Patent Owner disputes Petitioner's positions and argues that the challenged claims have not been shown to be unpatentable. We have already addressed Patent Owner's arguments relating to objective indicia of non-obviousness above, concluded that they are entitled to little weight, and will not revisit the issue. We review Patent Owner's other arguments below.

Patent Owner first argues that the Lenhard-Vilkin-Itzkowitz combination does not teach or suggest the claimed invention (however, this and each of Patent Owner's arguments appears to relate more to whether motivation to combine the prior art existed). Resp. 22–24 (citing Ex. 1004, 143; Ex. 1005, 2521; Ex. 1006, 112–13; Ex. 2022 ¶¶ 56, 69–70, 76, 112, 114).[14] Patent Owner argues that the Petitioner "picks and chooses" from Vilkin and Itzkowitz impermissibly to infer a motivation to modify Lenhard and that only "impermissible hindsight" would have led the ordinarily skilled artisan to make such changes to Lenhard. *Id.* at 22–23. Patent Owner argues Lenhard's samples were frozen and the reference does not provide details on sample collection, that Vilkin's methods require refrigeration, and Itzkowitz does not disclose separation of its sample and does disclose that patients prefer DNA testing to FOBT. *Id.* at 22–24.

[14] Patent Owner cites the added pagination of Petitioner's exhibits. Petitioner's briefing cites the references' original page numbers, as did our Institution Decision. Because of the burden of continuously translating Patent Owner's citations, we provide these initial cites to specific pages of the evidence in addressing Patent Owner's argument, but may not repeat it.

Patent Owner continues, arguing that Petitioner fails to establish that an ordinarily skilled artisan would have been motivated to combine the references, presenting several arguments. Resp. 24–37 (citing Ex. 1004, 143–44, 146, 148, Tables 2, 4; Ex. 1005, 2519–20, 2522–24; Ex. 1006, 116–117; Ex. 1013, 1576; Ex. 1024, 29; Ex. 2022 ¶¶ 29, 61–62, 67, 95–101, 103–119; Ex. 2027, 169:13–17). Patent Owner's first point is that Petitioner offers no explanation as to why the ordinarily skilled artisan would make CRC screening "more complex" for the sake of improving sensitivity by following Lenhard's teaching of combining DNA and FOBT testing, rather than trying to improve either style of test individually. *Id.* at 25–26. Patent Owner also argues that the combination test of Lenhard did not improve screening adenomas and that Lenhard itself identified other markers than its HIC1 as more sensitive (i.e., the SFRP2 marker). *Id.* at 26–27. Patent Owner argues that each of Vilkin's and Itzkowitz's test were more sensitive than Lenhard's combination test, suggesting that individual test styles would be preferred. *Id.* at 27–29. Relating to these arguments, Patent Owner also argues that Petitioner's proposed combination would have been "highly unpredictable," because combinations of biomarkers may perform worse than (or show no improvement over) using individual markers. *Id.* at 29–30.

Patent Owner argues that, considering Lenhard as a whole, the reference would have dissuaded the person of ordinary skill in the art from following its combo-screening test because, although increased sensitivity was achieved, there was also increased complexity and decreased specificity. Resp. 30–32. Further, the two-test technique of Lenhard did not improve detection of adenomas. *Id.* at 32.

Patent Owner also argues that the ordinarily skilled artisan would have been dissuaded from using Vilkin's techniques because of the required multi-day collection steps and the need for refrigeration until shipping to a lab. *Id.* at 32–34.

Patent Owner also argues that the combination of tests upon combining the prior art would have increased complexity (a repeated point) and costs, which would have de-motivated the artisan from following Petitioner's proposed course of prior art combination. *Id.* at 34–37.

Patent Owner argues that increasing patient interaction with a fecal sample, as per Petitioner's proposed combination of the prior art, would have de-motivated the ordinarily skilled artisan. Resp. 37–39. Patent Owner argues that Lenhard does not teach that the patient separates the sample, but only that the sample is separated, that Vilkin does not specify what constitutes its Hb stabilizing buffer, and Itzkowitz does not identify its DNA-stabilizing buffer. *Id.* at 37–39.

Patent Owner concludes with the argument that the person of ordinary skill in the art, rather than separating and respectively buffering two portions of a fecal sample, would have instead not separated the sample and used a single stabilizing buffer to preserve both nucleic acid and blood protein. *Id.* at 39.

Although Patent Owner predominantly argues over Grounds 1–3 and claims 1–20 as a group, Patent Owner provides separate arguments for Ground 2, taking the position that Kanaoka's results using COX-2 as a biomarker, with iFOBT, for CRC screening were not sufficiently strong so that an ordinarily skilled artisan would have considered following Kanaoka. *See* Resp. 49 (citing, *inter alia*, Ex. 1007 ¶¶ 65–67); Sur-reply 18.

3. Analysis

Although Patent Owner's arguments most heavily attack whether the ordinarily skilled artisan would have been motivated, i.e., had a reason, to combine Lenhard, Vilkin, and Itzkowitz, initially Patent Owner asserts that this prior art combination fails to teach the entirety of the subject matter of independent claim 1. *See* Resp. 22–24. We conclude that Petitioner has shown that the combination of Lenhard, Vilkin, and Itzkowitz teaches or suggests the entirety of the invention defined by the challenged claims. Because Patent Owner focuses its arguments only on the broadest invention covered by independent claim 1 and does not argue over Petitioner's proof as to the dependent claims, we, too, focus on claim 1. We adopt Petitioner's arguments and evidence as to claims 2–20 under Grounds 1–3. *See supra* Section II.F.1 (review of Petitioner's assertions regarding dependent claims).

Claim 1 is reproduced above (*see supra* Sections I.C (reproduced in full), II.F.1 (reproduced in a series of quoted limitations)), but, to summarize, it requires a method of processing "freshly-collected" and unfrozen fecal matter by collecting it "at home," by defecation into a sealable vessel, removing a portion of the sample and combining it with a buffer to preserve blood proteins, and combining the other portion of the sample with a stabilizing buffer (later claimed to preserve nucleic acids, e.g., DNA or RNA). Ex. 1001, 45:21–47:4. As described and claimed in the '781 patent, this method is used to detect blood proteins and/or target epigenetically modified DNA (or RNA), which, as discussed above, may detect colorectal cancer (CRC), although this ultimate objective or result is not required by any challenged claim. *See id.* at 45:39–67 (claims 2–20).

Petitioner's evidence shows that, before the priority date of the challenged claims, CRC screening by processing two portions of a human fecal sample to detect the presence of blood and/or epigenetically modified DNA (isolated, bisulfite-treated, purified, and PCR-amplified) was taught by Lenhard, which disclosed that a combined gFOBT+hypermethylated HIC1 detection method worked well to detect CRCs; and "[t]he combination of both assays resulted in increased detection rates for CRCs." Ex. 1004, 142–43, 145–48; *see also* Ex. 1002 ¶¶ 109–115, 139–142 (discussing Lenhard). Other possible genetic markers for CRC were also identified by Lenhard; e.g., hypermethylation at SFRP2, SFRP5, PGR, hMLH1, and HPP1, or other methylation markers, possibly used in combination with one another. *Id.* at 143, 148.

Lenhard unquestionably teaches a combined blood and nucleic acid, dual-portioned, fecal screening test for CRCs, but it is by no means the only incidence or suggestion of such in the prior art of record.

In 2001, Nishikawa disclosed a CRC screening method using DNA extracted from stool to detect mutations that may identify cancer, compared it to FOBT screening of the sample, and concluded that the DNA testing was "highly specific, noninvasive, and cost-effective, and ***should provide a more sensitive and specific tool for mass screening of colorectal cancer*** than is currently available, ***especially if used in combination with fecal occult blood testing***." Ex. 1011, 107–08, 112 (emphasis added).

In 2005, Kutzner disclosed that a combined FOBT and genetic screen (called "molecular diagnosis (MD)") for target DNA alterations was effective to screen for CRC and "[t]he ***combined application*** of the two methods together led to a ***significantly higher sensitivity*** than the application of any of the methods alone," and "suggest[ed] that ***MD and FOBT should complement rather than replace each other***." Ex. 1012, 33–37, 40 (emphasis added).

And, in 2008, Levin disclosed that the "range of options for CRC screening" included "2 general categories: stool tests, which include tests for occult blood or exfoliated DNA," such as guaiac-based fecal occult blood tests and fecal immunological tests (respectively, gFOBT and FIT), and structural exams, which include, *inter alia*, "colonoscopy (CSPY)," and further disclosed that "[t]hese ***tests may be used alone or in combination***," that "[c]ollection kits [for FOBT and DNA testing] ha[d] been designed to facilitate specimen collection and mailing and to enhance compliance," and reported that data showed "***combined results from a standard gFOBT and a panel of DNA markers*** (APC, BAT26, and L-DNA) resulted in a ***combined sensitivity for cancer of 93% and specificity of 89%***." Ex. 1013, 1570–78 (emphasis added).

None of Nishikawa, Levin, or Kutzner is asserted by Petitioner here against the claims as prior art, but each supports Petitioner's position, as explained by Petitioner's declarant, that, before the priority date of the '781 patent, there was nothing new or non-obvious about CRC screening by detecting both blood and DNA markers in a respectively portioned fecal sample, as taught by Lenhard. *See* Ex. 1002 ¶¶ 32, 49, 60, 62–65, 98, 241, 243–244 (discussing Nishikawa, Levin, and Kutzner).

As noted above, Lenhard disclosed the detection of blood in fecal samples using gFOBT (a blue color reaction indicated a positive result). Ex. 1004. Vilkin teaches that, as of 2005, there was a better test for blood in fecal samples—iFOBT (immunochemical detection of Hb). Ex. 1005, 2519.

Vilkin discloses that it was known that gFOBT had "low sensitivity for . . . CRC . . . and low specificity due to nonspecificity for human hemoglobin"; it required "diet restrictions" as well. *Id.* Vilkin explains that the iFOBT screen overcame these drawbacks and also offered quantitative results whereby a threshold could be set for colonoscopy. *Id.* Petitioner combines Vilkin's teachings with Lenhard's for this subject matter, substituting Lenhard's gFOBT screen with Vilkin's better iFOBT screen, and with it Vilkin's associated disclosed sampling device (tube) for (use at home in) portioning, storing, and preserving in an Hb stabilizing buffer, the fecal sample portion to be so tested, which did not require freezing, but only used refrigeration until its return to a lab for testing. *Id.* at 2520. With such a modification to Lenhard's methods and materials, most of the required steps of claim 1 are taught and obvious (i.e., the claim's preamble, the collection of a sample at home, removing a portion of the sample and combining it with a blood-protein-preserving buffer).

As noted above, Lenhard teaches screening of epigenetically modified nucleic acids, but does not mention sample-preserving with a stabilizing buffer. For this, Petitioner points to the teachings of Itzkowitz, which discloses that the immediate addition of a DNA preservation buffer to a fecal sample is advantageous because it prevents sample degradation for several days, even at room temperature, and allows higher screening sensitivity—better for detecting cancer—and the test had "a very high degree of patient satisfaction." Ex. 1006, 111–17. Moreover, the patients providing samples in Itzkowitz were given "a special stool collection kit for mounting to a toilet," where the buffer was immediately added to the collected sample, and then the samples were "shipped at room temperature overnight" to a clinical lab. *Id.* at 112. Similar to Vilkin's teachings, this suggests providing specimens at home directly in the same sealable container in which the sample is shipped. By modifying Lenhard's methods and materials to add Itzkowitz's nucleic acid preserving buffer, and collection materials and techniques, all steps and elements of claim 1 are addressed in the prior art.

Based on the above, we find that the combination of Lenhard, Vilkin, and Itzkowitz teaches or suggests each step and element of claims 1–9, 11, and 14–20, and that the addition to this prior art combination of the subject matter taught by Kanaoka and Derks teaches or suggests the subject matter of claims 10, 12, and 13. Contrary to Patent Owner's argument, we find no unreasonable "picking and choosing" in Petitioner's citations to the teachings of Vilkin and Itzkowitz because the subject matter they disclose and that would be incorporated into the methods of Lenhard is the substantial focus of each reference rather than a mere component that might be selected from a great number of options. *See* Ex. 1005; Ex. 1006. Therefore, we move on to analyze whether the person of ordinary skill in the art would have had reason or motivation to combine the prior art.

As an initial matter, Lenhard, Vilkin, and Itzkowitz, as well as Kanaoka and Derks, are each directed to and disclose methods of screening for colorectal cancer by detecting a marker in human feces, which may be blood or certain nucleic acids. *See* Exs. 1004–1008. Thus, each reference is from the same field of endeavor as the claimed invention, and the asserted prior art is analogous art. *In re Bigio*, 381 F.3d 1320, 1325 (Fed. Cir. 2004) ("Two separate tests define the scope of analogous prior art: (1) whether the art is from the same field of endeavor, regardless of the problem addressed and, (2) if the reference is not within the field of the inventor's endeavor, whether the reference still is reasonably pertinent to the particular problem with which the inventor is involved.”). Because the asserted prior art is analogous art, we find it is appropriately combined as asserted.

As noted above, Patent Owner argues there was no reason to combine the references because of impermissible hindsight, increased complexity in the combined prior art, possible teaching away, unpredictability, and potential patient dissatisfaction with a screening process bringing them into proximity with their fecal samples. *See supra* Section II.F.2 (summary of Patent Owner’s positions). Patent Owner also argues that the ordinarily skilled artisan would not have used two sample preserving buffers, as claimed, but just one. *See, e.g.*, Hr’g Tr. 42:23–43:10. We are not persuaded by any of these arguments.

As we discuss above, the concept of combining a test for the presence of blood in feces to screen for CRC with a test for the presence of cancer-indicative DNA (e.g., epigenetically altered, methylated) to screen for CRC was not a new one as of Lenhard’s disclosure in 2005. The record shows that researchers in the field agreed that this combination of tests would be advantageous. *See* Ex. 1004, 143, 147; Ex. 1011, 112; 1012, 33, 40; Ex. 1013, 1578; *see also* Ex. 1002 ¶¶ 110, 115, 139–141 (discussing the prior art’s disclosure of this combination of tests and related advantages, such as enhanced sensitivity to CRC, complementary results, decrease in false-negative results). But, as also discussed above, the method disclosed in Lenhard had room for improvement that was identified by Vilkin and Itzkowitz.

Lenhard’s disclosed blood screen test used gFOBT, which Vilkin identified as inferior to iFOBT because the internationally-accepted iFOBT screen was specific for human Hb, had sufficient (acceptable) specificity, better and quantitative sensitivity, used a convenient and self-contained sampling device (with included Hb stabilizing buffer), did not require freezing (used refrigeration), was stable for weeks, used a known and available automated desktop analyzer, could be used at home, and overcame the need for dietary restrictions of gFOBT. *See generally* Ex. 1005. We find these are very good reasons the ordinarily skilled artisan would have modified Lenhard's respective methods and materials to use those of Vilkin (i.e., a home-use, sealable container for a non-frozen portion of a fecal sample, containing a blood-preserving buffer, for an iFOBT test).[15] *See, e.g.*, Ex. 1002 ¶¶ 143–147 (discussing reasons to combine Vilkin and Lenhard).

Lenhard's disclosed hypermethylated DNA testing relies on immediate freezing of the fecal sample and requires a patient to visit a laboratory to provide the sample. Both are inconveniences that are remedied by Itzkowitz's teachings. *See* Ex. 1002 ¶ 152. Similar to Lenhard, Itzkowitz teaches fecal DNA (methylation) testing for CRC, but unlike Lenhard, Itzkowitz teaches that samples could be collected by a patient at home if they were provided a collection kit that could be mounted to a toilet and DNA-stabilizing buffer that could be added to preserve the sample, which could then be shipped at room temperature (overnight) to a test facility. *See* Ex. 1006, 112; *see also* Ex. 1002 ¶¶ 148–158 (discussing reasons to modify

[15] We note that Vilkin also expresses a need for an improved and more convenient, at-home, toilet-based stool-collecting and sample storage device or system to improve upon a zip-lock bag system. Ex. 1005, 2524. As discussed below, Itzkowitz meets this need.

Lenhard's methods with those of Itzkowitz). These are good reasons to modify Lenhard in view of Itzkowitz.

Upon combining the methods of Lenhard, Vilkin, and Itzkowitz, the ordinarily skilled artisan would advantageously achieve a home-test system having a first container for initially collecting (on a toilet) a feces sample, a secondary collection tube with human Hb stabilizing buffer to collect a smaller portion of the fecal sample for iFOBT testing, and a DNA stabilizing buffer to add to the main sample for preservation at room temperature shipping from which a methylated DNA screen may be conducted. "[I]n considering the disclosure of a reference, it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw therefrom." *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). In view of the prior art teachings, sample containers must be sealable in order to be packaged, shipped, and then tested (at least as sealable as Vilkin's capped storage tube and double zip-lock bags). Itkowitz teaches and suggests that such a system would meet with "a very high degree of patient satisfaction" and would likely "get patients screened who might otherwise avoid an invasive [CRC] screening test." Ex. 1006, 116–17.

Based on the above, we conclude there would have been significant motivation for the person of ordinary skill in the art to combine Lenhard, Vilkin, and Itzkowitz (and also Kanaoka and Derks for their limited purposes) and that Petitioner has identified a reason with rational underpinning why the ordinarily skilled artisan would have done so with a reasonable expectation of success.

We reject Patent Owner's argument that only impermissible hindsight would have led the ordinarily skilled artisan to this prior art combination. Of course, "[i]t is impermissible . . . simply to engage in a hindsight reconstruction of the claimed invention, using the [challenged patent's] structure as a template and selecting elements from references to fill the gaps." *In re Gorman*, 933 F.2d 982, 987 (Fed. Cir. 1991). But, where a skilled artisan would pursue "known options" from "a finite number of identified, predictable solutions," the resulting invention is obvious. *KSR*, 550 U.S. at 421. "Any judgment on obviousness is in a sense necessarily a reconstruction based upon hindsight reasoning, but so long as it takes into account only knowledge which was within the level of ordinary skill at the time the claimed invention was made and does not include knowledge gleaned only from applicant's disclosure, such a reconstruction is proper." *In re McLaughlin*, 443 F.2d 1392, 1395 (CCPA 1971). Such is the case here because Lenhard, Vilkin, and Itzkowitz (and Kanaoka and Derks) disclose the claimed steps and elements to have been advantageously used for the same purposes as the invention; no hindsight reconstruction was required.

As to Patent Owner's arguments directed to potential increased complexity of the combined prior art process compared to any individual test, possible unpredictability, or possible patient dissatisfaction, we are not persuaded. The cited prior art, and even the '781 patent itself, expresses that screening tests like those of the Lenhard, Vilkin, and Itzkowitz (ultimately, iFOBT and methylated DNA detection) were well known. *See* Ex. 1001, 2:40–3:67 (background); Ex. 1004–1013. There is no persuasive evidence of unpredictability, and some of the cited (and asserted) prior art supports the fact that prior art techniques led to high patient satisfaction, and some were already in wide use internationally. *See, e.g.*, Ex. 1005; Ex. 1006.

Regarding Patent Owner's argument that a POSA "would not have been motivated by the teachings of Lenhard to combine classes of markers" given where "the specific HIC1/gFOBT combination reduced specificity and that other methylation markers alone achieved greater sensitivity," where "the reported increase in sensitivity was only in detection of CRC," and where "where Lenhard's specific HIC1/gFOBT combination showed no improvement in detection of advanced adenomas," Lenhard indicated its detection method was "sensitive and specific" and "well suited for stool-based CRC screening" and that adding more markers could "allow for the highly sensitive and specific stool-based detection of CRCs and adenomas." Ex. 1004, 148. Moreover, Vilkin teaches that, by using its improved iFOBT method, one of ordinary skill can achieve a high sensitivity of 76.5% and an acceptable specificity of 95.3%. Ex. 1005, 2523. Finally, Itzkowitz explains that the improvements in sensitivity offered by its methods came "at the expense of a relatively minor decrease in specificity." Ex. 1006, 116. Thus, the prior art asserted here specifically addresses Patent Owner's identified concerns as acceptable minor detractors from the advantages of their methods. *See Medichem S.A. v. Rolabo S.L.*, 437 F.3d 1157, 1165 (Fed. Cir. 2006) ("The fact that the motivating benefit comes at the expense of another benefit, however, should not nullify its use as a basis to modify the disclosure of one reference with the teachings of another. Instead, the benefits, both lost and gained, should be weighed against one another.").

As to Patent Owner's argument that the two-buffer system of the invention is non-obvious because the ordinarily skilled artisan would have preferred a single preserving buffer system to using two buffers, we are unpersuaded. There is no evidence in the record that a single, useful, preserving buffer for both blood proteins and nucleic acids was known to exist, much less one to be used for CRC screening fecal samples. On this record, it does not appear that this alternative was an option for the ordinarily skilled artisan.

Finally, as to Patent Owner's argument (directed to claims 12 and 13) that Kanaoka's results from using COX-2 as a biomarker, along with iFOBT, for CRC screening were not sufficiently strong to persuade an ordinarily skilled artisan to have considered using COX-2, we are not persuaded. As Dr. Whitney identifies, adding another biomarker to the Lenhard-Vilkin-Itzkowitz combined teachings,[16] specifically COX-2 RNA, would have been attractive to the ordinarily skilled artisan because Kanaoka discloses its process works (high sensitivity and specificity) and also that COX-2 was effectively combined with iFOBT in a complementary way (each detects certain CRC the other does not). Ex. 1002 ¶¶ 225–226; Ex. 1007 ¶¶ 55–68. Moreover, Kanaoka states that "the *detection of COX-2* from the feces by the RT-PCR technique had 90% sensitivity and 100% specificity, and it is *proved that the present invention is superior* to a conventional technique of the immunological fecal occult blood test," which suggests using such a technique with the methods of the other asserted prior art. Ex. 1007 ¶ 75 (emphasis added).

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[16] Note, Lenhard (Ex. 1004, 143, 147–48) suggests using other biomarkers (e.g., SFRP2, SFRP5, PGR were known) in addition to methylated HIC1, and Itzkowitz (Ex. 1006) teaches targeting two genetic biomarkers—e.g., "DY" and vimentin.

4. Summary and Conclusion

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 1–9, 11, and 14–20 are unpatentable as obvious over Lenhard, Vilkin, and Itzkowitz, that claims 12 and 13 are unpatentable as obvious over this same prior art combination adding Kanaoka's disclosed techniques of using RNA as a CRC biomarker, and that claim 10 is unpatentable as obvious over the first prior art combination adding Derks's disclosure of using (methylated) GATA-4 as a biomarker for CRC screening.

G. OBVIOUSNESS OVER SHUBER AND VILKIN, ALSO COMBINED WITH KANAOKA OR DERKS (GROUNDS 4, 5, AND 6)

We conclude above that all claims have been proven by a preponderance of the evidence to be unpatentable under Grounds 1–3, but for completeness, we review and analyze Petitioner's Grounds 4–6 below.

Under Ground 4, Petitioner asserts that claims 1–9, 11, and 14–20 would have been obvious over Shuber and Vilkin (asserted similarly as it was applied under Ground 1) and are, therefore, unpatentable. Pet. 47–62 (citing Ex. 1001, 47–49 (addressing hypermethylation); Ex. 1002 ¶¶ 180, 213, 235–249, 251–264, 266–276, 278–288; Ex. 1004, 143, 147; Ex. 1005, 2519–22, Table 1; Ex. 1009, 7:10–14, 7:19–21, 10:10–15, 14:1–5, 15:18–16:17, 29:8–19; Ex. 1011, 112; Ex. 1012, 34; 1013, 1578). Under Ground 5, Petitioner adds Kanaoka to the Shuber-Vilkin combination, similarly to how the reference was applied to the Lenhard-Vilkin-Itzkowitz combination under Ground 2, and asserts that claims 12 and 13 would have been obvious over this combination of prior art. *Id.* at 62–64 (citing Ex. 1002 ¶¶ 289–292, 294; Ex. 1007 ¶¶ 21–26, 47–48, 55–68, 76; Ex. 1030, 131). And, under Ground 6, Petitioner adds Derks to the Shuber-Vilkin combination, applied as it was under Ground 3, asserting claim 10 would have been obvious over this combination of prior art. Pet. 65 (citing Ex. 1002 ¶¶ 295–297; Ex. 1008, 250, 252, Table 3b; Ex. 1009, 7:19–21).

Patent Owner opposes, focusing on Petitioner's Ground 4 and, again, arguing over the invention defined by independent claim 1 of the '781 patent. Resp. 39–48 (citing Ex. 1001, 45:23–29; Ex. 1005, 2520, 2522; Ex. 1009, 14:1–16, 28:18–27, 29:8–19; Ex. 1011, 112; Ex. 1012, 40; 1013, 1578; Ex. 2022 ¶¶ 67, 69–73, 89–93, 95–107, 120–136).[17] As it did for Grounds 2 and 3, Patent Owner argues Grounds 5 and 6 together as a group with Ground 4, indicating that they fall (or, conversely, stand) with Ground 4, and also asserting that, as discussed above, Kanaoka (Ground 5) would not have motivated the ordinarily skilled artisan to combine the prior art to achieve the invention of claims 12 and 13, but, rather, would have discouraged them. *Id.* at 49 (citing Ex. 1007 ¶¶ 65–67; Ex. 2022 ¶ 138).

1. Petitioner's Positions

Under Ground 4, Petitioner asserts that Shuber discloses CRC screening of a fecal sample, provided by a patient at home, directly in a container for shipping, and preserved therein using a stabilizing solution, without requiring freezing, using DNA analysis to detect hypermethylation. Pet. 48 (citing Ex. 1002 ¶¶ 235–245; Ex. 1009, 7:12, 7:19–21, 10:12–14, 29:8–19). Similar to Ground 1, Petitioner asserts Vilkin would be combined with Shuber for its teaching of a quantitative iFOBT CRC screen, and a sealable container and stabilizing buffer for the sample, which would be

[17] As noted above, for some references, Patent Owner cites the added, rather than the original pagination. As has Petitioner, we cite original pagination.

combined with Shuber's DNA screen as a complementary test. *Id.* at 47 (citing Ex. 1005, 2520–21).

Petitioner recounts that, before the priority date of the '781 patent, it was understood in the relevant art that combining a fecal DNA test and an FOBT test had advantages in CRC screening. *Id.* at 48–50 (citing Ex. 1004, 143, 147; Ex. 1011, 112; Ex. 1012, 34; Ex. 1013, 1578). Petitioner asserts that the person of ordinary skill in the art would have been motivated to combine the CRC screening methods of Shuber and Vilkin for such advantages, with a reasonable expectation of success. *Id.* at 50–51 (citing Ex. 1002 ¶¶ 240–246; Ex. 1005, 2520; Ex. 1009, 10:10–15, 29:8–19).

Petitioner also maps the disclosures of Shuber and Vilkin to the steps and elements of the challenged claims, adding Kanaoka or Derks to address dependent claims 10, 12, and 13. *Id.* at 51–65.

Regarding claim 1's preamble ("A method of processing a freshly-collected fecal sample without freezing"), Petitioner asserts that both of Shuber and Vilkin teach that their samples are "freshly-collected" and not frozen. *Id.* at 51 (citing Ex. 1002 ¶¶ 248–249; Ex. 1005, 2520; Ex. 1009, 7:10–12).

Regarding claim 1's first recited step (a) ("collecting a fecal sample from a human subject, wherein the fecal sample is collected at home by the human subject by defecation directly into a sealable collection vessel"), Petitioner asserts it is taught by Shuber's disclosure of sample collection by direct defecation into a sealable container (sized and configured for sealing so that it may be shipped), where the patient may add (or the container may already contain) stabilizing solution, which suggests this collection is performed at home. *Id.* at 51–52 (citing Ex. 1002 ¶¶ 252–253, 251–254; Ex. 1009, 10:12–14, 29:8–19).

Petitioner addresses claim 1's next two recited steps (b) ("removing a portion of the fecal sample to a separate sealable container to produce a removed portion and a remaining portion of the fecal sample" and (c) ("combining the removed portion of the fecal sample in the separate sealable container with a buffer that prevents denaturation or degradation of blood proteins found in a fecal sample, and sealing the sealable container"), again asserting that they are taught by Vilkin in the same fashion as discussed above regarding Ground 1. *Id.* at 52–53 (citing Ex. 1002 ¶¶ 255–258; Ex. 1005, 2520, Fig. 1); *see also supra* Section II.F.

Regarding the final required step (d) of claim 1 ("combining the remaining portion of the fecal sample in the sealable collection vessel with a stabilizing buffer, and sealing the collection vessel"), Petitioner asserts Shuber teaches this in disclosing the stool sample can be "directly deposited into a container (e.g., a sealable container)" to which the "stabilizing solution may be added," and the "container with sample and stabilization solution may be sealed for storage/shipping." Pet. 54–55 (citing Ex. 1001 ¶¶ 259–261; Ex. 1009, 7:19–21, 29:8-19).

As was the case with Grounds 1–3, Petitioner also addresses each dependent claim as taught or suggested by Shuber and Vilkin, adding Kanaoka or Derks regarding claims 10, 12, and 13.

Regarding claim 2, which requires "delivering [the portioned sample] to a medical diagnostics laboratory," Petitioner asserts the prior art combination teaches this because each of Shuber and Vilkin discloses delivering fecal samples to a laboratory for testing. Pet. 55 (citing Ex. 1002

¶¶ 262–264; Ex. 1005, 2520; Ex. 1009, 10:10–15). This factual assertion is not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

Claim 3, directed to a method of processing the portioned sample of claim 1 by testing the first sample portion "for an amount of blood protein" and testing the second sample portion, after "extracting nucleic acid," for "an amount of human nucleic acid," which Petitioner asserts is taught by Vilkin's iFOBT testing of fecal samples, and Shuber's isolating and purifying DNA and DNA testing of fecal samples. Pet. 55–56 (citing Ex. 1002 ¶¶ 266–271; Ex. 1005, 2519–20, 2522, Table 1; Ex. 1009, 7:19–21, 10:10–15, 15:18–16:17). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

Regarding claim 4, which requires "determining expression from a human gene" when testing for the nucleic acid of claim 3, claim 5, which requires that such expression comprises identifying "an epigenic modification," and claim 6, which requires "measuring an amount of a methylated human DNA," Petitioner asserts that Shuber teaches this claimed subject matter in disclosing testing for hypermethylation of HLTF and V29 loci in fecal samples. Pet. 58–59 (citing Ex. 1002 ¶¶ 272–275; Ex. 1009, 14:1–5). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

Regarding claims 7, 8, and 9, which respectively require that the epigenetic modification analyzed "comprises aberrant methylation," that the "aberrant methylation comprises hypermethylation," and that the human DNA "comprises a gene and/or a promoter region of a gene," Petitioner asserts this is taught by Shuber's disclosing testing fecal samples for hypermethylated human genes. Pet. 59–60 (citing Ex. 1001, 12:47–49 (addressing hypermethylation); Ex. 1002 ¶¶ 277–279; Ex. 1009, 14:1–5). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

Regarding claim 11, requiring "modifying the nucleic acid with bisulfate ions under conditions where[] unmethylated cytosine is converted to uracil," Petitioner again points to Shuber's disclosure of testing for gene methylation, which, according to Petitioner, uses bisulfite conversion as a standard, where bisulfite ions are used to convert unmethylated cytosine to uracil (asserting the claim's reference to "bisulfate" instead of "bisulfite" is a typographical error). Pet. 60–61 (citing Ex. 1002 ¶¶ 180, 213, 280–282; Ex. 1009, 14:1–5). (These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18).

Claim 14 requires that, in the method of claim 3, testing for blood protein "comprises testing for a concentration of hemoglobin," and claim 15 requires that this testing for hemoglobin "comprises immunochemical detection of hemoglobin," which Petitioner asserts is taught by Vilkin in disclosing iFOBT, which (quantitatively) tests for hemoglobin. Pet. 61 (citing Ex. 1002 ¶¶ 283–284; Ex. 1005, 2519–22, Table 1). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

Regarding claims 16–20, each requires detecting a certain threshold of hemoglobin in the fecal sample, i.e., at least 5, 10, 20, 50, or 200 ng/ml, which Petitioner asserts would have been obvious over the prior art combination because these thresholds amount to routine optimization of a result-effective variable; Vilkin explains that the physician can choose an optimal fecal hemoglobin threshold to trigger a follow-up colonoscopy. Pet. 61–62 (citing Ex. 1002 ¶¶ 285–288). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

Regarding claims 12 and 13, Petitioner adds the teachings of Kanaoka to those of Shuber and Vilkin in asserting obviousness. Claim 12 requires measuring RNA expressed from a gene and claim 13 requires that the RNA is measured using reverse transcriptase polymerase chain reaction (RT-PCR), each of which Petitioner asserts is taught by Kanaoka in disclosing a COX-2 RNA biomarker detection process, using RT-PCR, which would be combined with Shuber and Vilkin and used with a reasonable expectation of success because Kanaoka's successful working example "has a higher detection sensitivity," and would complement the other references' detection processes. Pet. 62–64 (citing Ex. 1002 ¶¶ 289–294; Ex. 1007 ¶¶ 21–26, 47–48, 55–68, 76; Ex. 1030, 131).

Regarding claim 10, Petitioner adds the teachings of Derks to those of Shuber and Vilkin in asserting obviousness. Claim 10 requires that the gene having an epigenetic modification be selected from the group consisting of PHACTR3, NDRG4, FOXE1, GATA4, GPNMB, TFP12, SOX17, SYNE1, LAMA, MMP2, OSMR, SFRP2, and CDO1, and Petitioner asserts this is taught by Derks in disclosing testing for GATA-4, which Petitioner asserts would be combined by the ordinarily skilled artisan as an additional marker with Shuber's and Vilkin's methods, to improve sensitivity of screening, with a reasonable expectation of success. *Id.* at 46–47 (citing Ex. 1002 ¶¶ 295–297; Ex. 1008, 250, 252, Table 3b). These factual assertions are not specifically disputed by Patent Owner. *See* Resp. 39–49; Sur-reply 14–18.

2. Patent Owner's Positions

Patent Owner argues that Shuber "says little about sample collection and nothing about at-home collection by a patient," and that Vilkin does not identify where the target stool is when a portion is collected with a probe. *Id.* at 41–42 (citing Ex. 1001, 45:23–29; Ex. 1005, 2020; Ex. 1009, 29:8–19; Ex. 2022 ¶¶ 69-73, 89–93).

As with Grounds 1–3, most of Patent Owner's opposition is directed to a lack of reason to combine the prior art. Patent Owner argues that logic suggests that Shuber and Vilkin, considering the widely recognized (at the time) need for improved CRC screening, would only be combined with the benefit of hindsight. Resp. 40. Patent Owner argues that Shuber does not suggest combining its methods with any other CRC screening tests than nucleic acid-based tests; not a test that detects blood or any other category. *Id.* at 40–41 (citing Ex. 1009, 14:1–15, 28:18–27; Ex. 2022 ¶ 90). Patent Owner argues that, because Vilkin discloses stool testing over three days, this would dissuade a person from using such a test. *Id.* at 41, 47 (citing Ex. 1005, 2522). Patent Owner argues that Shuber discloses that its methods were so sensitive that there would have been no reason for the person of ordinary skill in the art to have added Vilkin's iFOBT testing, as doing so would have likely not improved sensitivity, but would have reduced specificity and been more complicated. *Id.* at 46 (citing Ex. 1009, 14:15–16; Ex. 2022 ¶¶ 120–134). Patent Owner argues that there also would have been no motivation to combine Shuber and Vilkin because doing so would have increased the cost and complexity of the testing. *Id.* at 47–48 (citing Ex. 2022 ¶ 134).

Further to the arguments identified above, Patent Owner argues that the ordinarily skilled artisan would not have been motivated to combine Shuber and Vilkin because their combination does not suggest enhanced performance, and the supporting prior art references (Nishikawa, Lenhard, Kutzner, and Levin) are mere speculation and are inconclusive that combining DNA and blood testing for CRC screening would be advantageous. Pet. 42–48 (citing Ex. 1004, 6; Ex. 1011, 6; Ex. 1012, 8; Ex. 1013, 9; Ex. 2022 ¶¶ 67, 95–107, 120–136).

As for Grounds 5 and 6, Patent Owner argues that they fail for the same reasons as for Ground 3. *Id.* at 49. Regarding claims 12 and 13 and Ground 5, again Patent Owner argues that Kanaoka's results were not good enough to have added its COX-2 (RNA) as a marker for screening. *Id.* (citing Ex. 1007 ¶¶ 65–68; Ex. 2022 ¶ 138).

3. Analysis

Again, Patent Owner focuses its arguments on the invention of challenged claim 1; we do the same. We adopt Petitioner's arguments and evidence as to claims 2–20 under Grounds 4–6, as well. *See supra* Section II.G.1 (review of Petitioner's assertions regarding dependent claims). We conclude that Shuber and Vilkin would have been combined by the person of ordinary skill in the art, with a reasonable expectation of success, and that doing so would have achieved the invention of the challenged claims. We explain our reasoning and address Patent Owner's arguments below.

Claim 1 is reproduced above (*see supra* Sections I.C (reproduced in full), G.1 (reproduced in a series of quoted limitations), but to again summarize, it requires a method of processing "freshly-collected" and unfrozen fecal matter by collecting it "at home," by defecation into a sealable vessel, removing a portion of the sample and combining it with a buffer to preserve blood proteins, and combining the other portion of the sample with a stabilizing buffer (later claimed to preserve nucleic acids, e.g., DNA or RNA). Ex. 1001, 45:21–47:4. As described and claimed in the ’781 patent, this method is used to detect blood proteins and/or target epigenetically modified DNA (or RNA), which, as discussed above, may detect colorectal cancer (which detection is not required by any challenged claim). *See id.* at 45:39–67 (claims 2–20).

We have above already analyzed the teachings of Vilkin in addressing Grounds 1–3 and the same analysis applies to Petitioner’s Grounds 4–6—in summary, Vilkin discloses an iFOBT sample collection kit comprising a sealable container with Hb preserving buffer that overcomes many of the drawbacks of the prior art and a related method for CRC screening. *See supra* Section II.F.3.

Before the priority date of the challenged claims, Shuber shows it was known in the relevant art to screen for CRC (carcinoma or adenoma) by assaying a fecal sample for nucleic acid integrity or mutation and to stabilize the nucleic acids in the sample with a buffer in preparation for the assay. Ex. 1009, Abstract, 1:11–13, 2:5–9, 2:27–32, 4:14–16, 6:9–11, 9:20–22, 23:12–25:2. Shuber further teaches that such a stool sample is not frozen, and may be exposed to a variety of temperatures during transport or storage and, because of the nucleic acid stabilizing solution, may be collected at a remote location and mailed to a testing center. *Id.* at 3:1–22, 7:10–12, 10:10–15. Shuber teaches that patients “directly deposited” their stool sample “into a . . . sealable container[] that already contains stabilization solution,” or the “stabilization solution may be added to a container” after the sample is deposited therein, which is then "sealed for storage/shipping," and "smaller or larger containers may be used." *Id.* at 29:8–19.

As discussed above, the evidence of record (e.g., Lenhard, Nishikawa, Kutzner, and Levin) also supports that the combined application of an epigenetically modified genetic assay and fecal blood assay screening methods, used together, led to a significantly higher sensitivity than either alone, and that iFOBT and DNA/RNA marker assays would complement rather than replace each other. *See* Ex. 1004; Ex. 1011; Ex. 1012; Ex. 1013. Shuber, like Vilkin, is directed to and discloses methods of screening for colorectal cancer by detecting a marker in human feces. *See* Ex. 1005; Ex. 1009. Thus, Shuber, like Vilkin, is from the same field of endeavor as the claimed invention and is analogous art. *In re Bigio*, 381 F.3d at 1325. The ordinarily skilled artisan would have had good reason to combine the methods of Shuber and Vilkin to achieve a better CRC screening method than either method provided individually.

When combined, Shuber and Vilkin teach or suggest each step and element of independent claim 1, and of dependent claims 2–20 (adding the relevant disclosures of Kanaoka and Derks). As discussed above, Shuber and Vilkin together teach processing "freshly-collected" and unfrozen fecal matter by collecting it "at home," by defecation into a sealable vessel, removing a portion of the sample and combining it with a buffer (in another sealable container) to preserve blood proteins, and combining the other portion of the sample with a stabilizing buffer (later claimed to preserve nucleic acids, e.g., DNA or RNA), to then detect blood proteins and/or target epigenetically modified DNA (or RNA), to detect colorectal cancer (CRC). *See* Ex. 1001, 45:21–47:4. Kanaoka and Derks each would have been combined with Shuber and Vilkin just as either would have been combined with Lenhard, Vilkin, and Itzkowitz, and the resultant teachings would be essentially the same. *See supra* Section II.F.

Because of the record's strong evidence (e.g., Lenhard, Nishikawa, Kutzner, and Levin) that the person of ordinary skill in the art would have had a reason to combine Shuber and Vilkin, with a rational underpinning, and would have reasonably expected success, we are not persuaded by any of Patent Owner's arguments that no motivation existed to do so. *See* Ex. 1004; Ex. 1011; Ex. 1012; Ex. 1013; *see also* Ex. 1002 ¶¶ 241–246.

Shuber, itself, may not express that its methods should have been combined with Vilkin's iFOBT screening method, but the other cited prior art identifies that this motivation was well established before the challenged claims. "[E]vidence of a motivation to combine need *not* be found in the prior art references themselves, but rather may be found in 'the knowledge of one of ordinary skill in the art." *Dystar Textilfarben Gmbh & Co. Deutschland KG v. C.H. Patrick Co.*, 464 F.3d 1356, 1366 (Fed. Cir. 2006). And, an obviousness "analysis 'need not seek out precise teachings directed to the specific subject matter of the challenged claim, [to] take account of the inferences and creative steps that a person of ordinary skill in the art would employ."' *Intel Corp. v. PACT XXP Schweiz AG*, 61 F.4th 1373, 1379–80 (Fed. Cir. 2023) (quoting *KSR*, 550 U.S. at 418, 420). Moreover, the prior art combination does not need to be the best or preferred approach but need only be "desirable in light of the prior art." *Honeywell Int'l Inc. v. 3G Licensing, S.A.*, 124 F.4th 1345, 1356 (Fed. Cir. 2025). Finally, the "case law is clear that obviousness cannot be avoided simply by a showing of some degree of unpredictability in the art so long as there was a reasonable probability of success." *Pfizer, Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1364 (Fed. Cir. 2007).

As for Patent Owner's argument that neither Shuber nor Vilkin explicitly discloses that a patient provides their sample at home, we are not persuaded. Each of Shuber's and Vilkin's collection systems are designed for easy patient use, i.e., Shuber's is a sealable and shippable container where the patient unleashes directly in the container and Vilkin's is a sealable tube with a poker-cap for collecting a small portion of feces, each of which is to be shipped by the patient to an analysis lab. Ex. 1005, 2520, Fig. 1; Ex. 1009, 29:8–19. Thus, we agree with Petitioner's declarant that the "at home" claim element is taught or suggested by and obvious over the prior art. *See* Ex. 1002 ¶¶ 251–254.

4. Summary and Conclusion

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 1–9, 11, and 14–20 are unpatentable as obvious over Shuber and Vilkin, claims 12 and 13 are unpatentable as obvious over Shuber, Vilkin, and Kanaoka, and claim 10 is unpatentable as obvious over Shuber, Vilkin, and Derks.

III. CONCLUSION[18]

On the record before us, Petitioner demonstrates by a preponderance of the evidence that claims 1–20 of the '781 patent would have been obvious

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[18] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application

over the asserted prior art, under each asserted ground. Accordingly, claims 1–20 of the ’781 patent are unpatentable under 35 U.S.C. § 103.

Our final decision is summarized as follows:

Claim(s)	**35 U.S.C. §**	**Reference(s)/ Basis**	**Claim(s) Shown Unpatentable**	**Claim(s) Not Shown Unpatentable**
1–9, 11, 14–20	103	Lenhard, Vilkin, Itzkowitz	1–9, 11, 14–20	
12, 13	103	Lenhard, Vilkin, Itzkowitz, Kanaoka	12, 13	
10	103	Lenhard, Vilkin, Itzkowitz, Derks	10	
1–9, 11, 14–20	103	Shuber, Vilkin	1–9, 11, 14–20	
12, 13	103	Shuber, Vilkin, Kanaoka	12, 13	
10	103	Shuber, Vilkin, Derks	10	
Overall Outcome			1–20	

or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

ORDER

Accordingly, it is hereby:

ORDERED that, pursuant to 35 U.S.C. § 314, per *inter partes* review of claims 1–20 of U.S. Patent 11,634,781 B2, the challenged claims are unpatentable; and

FURTHER ORDERED that, as this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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